Problem Set 1 – Complete Step-by-Step Solution Set

Statistical Learning / Regression / Classification

Full questions quoted from the uploaded assignment; theory is stated before each solution.

Source basis. This solution set uses the uploaded *Problem Set 3.pdf* (whose first page is headed "Problem Set 1") and the uploaded *SL_I_midsem.pdf.* The relevant course material covers least-squares regression, LAD regression, ridge/LASSO/elastic net, splines, perceptrons, Nadaraya–Watson regression, regression trees, impurity functions, and Bayes classification.

Contents

Question 1

Whole question from the assignment.

(a) Based on the following data set, find the product-moment correlation coefficient between $\log(X)$ and $\log(Y)$. [Hint: here the product of X and Y is always 144].

x	1	2	3	4	6	8	9	12
y	144	72	48	36	24	18	16	12

(b) For a bivariate data set, when do the two types of regression lines: (i) Y on X and (ii) X on Y become identical?

(c) Suppose that the equations of the two regression lines for a bivariate dataset are $2X + 3Y = 17$ and $3X + 4Y = 24$. Find the mean of X, the mean of Y, $\operatorname{corr}(X, Y)$ and $\operatorname{var}(X)/\operatorname{var}(Y)$.

Part (a)

Theory required for this solution. For two non-degenerate variables,

$$r_{UV} = \frac{\operatorname{Cov}(U, V)}{\sqrt{\operatorname{Var}(U)\operatorname{Var}(V)}}.$$

In particular, $r_{UV} = 1$ or -1 exactly when there is an exact affine relationship between U and V, with positive or negative slope respectively. If $XY = c > 0$ is constant, then

$$\log Y = \log c - \log X,$$

which is an exact affine relation with slope -1.

Step 1. From the hint, $XY = 144$ for every observation.

$$Y = \frac{144}{X}.$$

Taking logarithms,

$$\log Y = \log 144 - \log X.$$

Why this step leads to the next step: If we put $U = \log X$ and $V = \log Y$, then the displayed equation becomes $V = \log 144 - U$. Thus every point (U, V) lies exactly on the straight line with slope -1.

Step 2. Therefore the product-moment correlation is the correlation of two variables satisfying an exact negative linear relation:

$$r_{\log X, \log Y} = -1.$$

Final conclusion: $\boxed{\operatorname{Corr}(\log X, \log Y) = -1}$.

Part (b)

Theory required for this solution. For a bivariate sample, the regression coefficients are

$$b_{Y|X} = r\frac{s_Y}{s_X}, \qquad b_{X|Y} = r\frac{s_X}{s_Y}.$$

The two regression lines always pass through the sample mean point $(\bar{X}, \bar{Y})$. If the two lines are identical, their slopes (when written as Y against X) must agree. Equivalently,

$$b_{Y|X}b_{X|Y} = r^2.$$

Because identical inverse lines must have reciprocal slopes, the required condition is $r^2 = 1$.

Step 1. The line of Y on X has slope $b_{Y|X}$ and the line of X on Y has inverse slope $b_{X|Y}$.

Step 2. If the two geometric lines are the same, then the inverse slopes must satisfy

$$b_{Y|X}b_{X|Y} = 1.$$

Step 3. But

$$b_{Y|X}b_{X|Y} = \left(r\frac{s_Y}{s_X}\right)\left(r\frac{s_X}{s_Y}\right) = r^2.$$

Hence $r^2 = 1$.

Step 4. Conversely, if $r = \pm 1$, all sample points lie exactly on one straight line. The two regression lines then reduce to that same line.

Final conclusion: The two regression lines are identical exactly when

$$\boxed{|r| = 1}$$

for a non-degenerate bivariate data set.

Part (c)

Theory required for this solution. Both regression lines pass through $(\bar{X}, \bar{Y})$. Also, if a line is written as $Y = a + bX$, its slope is $b_{Y|X}$; if written as $X = c + dY$, its slope coefficient is $b_{X|Y}$. Finally,

$$b_{Y|X} = r\frac{s_Y}{s_X}, \qquad b_{X|Y} = r\frac{s_X}{s_Y}, \qquad b_{Y|X}b_{X|Y} = r^2.$$

Step 1. Since both lines pass through $(\bar{X}, \bar{Y})$, their common point is obtained by solving

$$2\bar{X} + 3\bar{Y} = 17, \qquad 3\bar{X} + 4\bar{Y} = 24.$$

Eliminate $\bar{X}$:

$$4(2\bar{X} + 3\bar{Y}) = 68, \qquad 3(3\bar{X} + 4\bar{Y}) = 72.$$

A quicker elimination gives

$$\bar{X} = 4, \qquad \bar{Y} = 3.$$

Why this step leads to the next step: A regression line is not an arbitrary fitted line: both regression lines pass through the sample-mean point. Therefore their intersection immediately gives the means.

Step 2. Rewrite the first line as a regression of Y on X:

$$2X + 3Y = 17 \iff Y = \frac{17}{3} - \frac{2}{3}X.$$

Thus

$$b_{Y|X} = -\frac{2}{3}.$$

Rewrite the second as a regression of X on Y:

$$3X + 4Y = 24 \iff X = 8 - \frac{4}{3}Y.$$

Thus

$$b_{X|Y} = -\frac{4}{3}.$$

Step 3. Their product is

$$r^2 = b_{Y|X} b_{X|Y} = \left(-\frac{2}{3}\right)\left(-\frac{4}{3}\right) = \frac{8}{9}.$$

Both slopes are negative, so $r < 0$. Therefore

$$r = -\sqrt{\frac{8}{9}} = -\frac{2\sqrt{2}}{3}.$$

Step 4. Use $b_{Y|X} = r(s_Y/s_X)$:

$$-\frac{2}{3} = -\frac{2\sqrt{2}}{3}\frac{s_Y}{s_X}.$$

Cancel the common negative factor:

$$1 = \sqrt{2}\frac{s_Y}{s_X} \quad \Longrightarrow \quad \frac{s_Y}{s_X} = \frac{1}{\sqrt{2}}.$$

Hence

$$\frac{\mathrm{Var}(X)}{\mathrm{Var}(Y)} = \frac{s_X^2}{s_Y^2} = 2.$$

Final conclusion:

$$\boxed{\bar{X} = 4, \quad \bar{Y} = 3, \quad r = -\frac{2\sqrt{2}}{3}, \quad \frac{\mathrm{Var}(X)}{\mathrm{Var}(Y)} = 2.}$$

Question 2

Whole question from the assignment.

(a) Consider the following data set consisting of 10 bivariate observations

x	0	1	2	3	4	5	6	7	8	9
y	1	3	5	8	11	13	17	19	23	30

Find the value of β that minimizes $\sum_{i=1}^{10} |y_i - \beta x_i|$. Comment on the uniqueness of this minimizer.

(b) Consider n bivariate observations $(x_1, y_1), \ldots, (x_n, y_n)$, where $x_i = 2^{i-1}$ for $i = 1, 2, \ldots, n$. Show that the minimizer of $\Delta(\beta) = \sum_{i=1}^{n} |y_i - \beta x_i|$ is y_n/x_n.

(c) If the mean and median of 10 univariate observations $x_1, \ldots, x_{10}$ are 2 and 3, respectively, show that the minimizer of

$$\psi(\theta) = \sum_{i=1}^{10} \{2(x_i - \theta)^2 + 3|x_i - \theta|\}$$

cannot lie outside the closed interval $[2, 3]$.

Part (a)

Theory required for this solution. A basic LAD fact from the course notes is:

$$\sum_{i=1}^{n} |z_i - \alpha| \quad \text{is minimized when } \alpha \text{ is a median of } z_1, \ldots, z_n.$$

More generally,

$$\sum_{i=1}^{n} |y_i - \beta x_i| = \sum_{i:x_i \neq 0} |x_i| \left| \frac{y_i}{x_i} - \beta \right| + \text{constant},$$

so the minimizer is a weighted median of the ratios y_i/x_i, with weights $|x_i|$.

Step 1. The observation with $x_1 = 0$ contributes

$$|y_1 - \beta x_1| = |1| = 1,$$

which does not depend on β. Ignore this constant for minimization.

Step 2. For the remaining observations compute $z_i = y_i/x_i$:

$$3, \frac{5}{2}, \frac{8}{3}, \frac{11}{4}, \frac{13}{5}, \frac{17}{6}, \frac{19}{7}, \frac{23}{8}, \frac{10}{3}.$$

The corresponding weights are

$$1, 2, 3, 4, 5, 6, 7, 8, 9.$$

Step 3. Sort the ratios:

$$\frac{5}{2}, \frac{13}{5}, \frac{8}{3}, \frac{19}{7}, \frac{11}{4}, \frac{17}{6}, \frac{23}{8}, 3, \frac{10}{3}.$$

Their weights are respectively

$$2, 5, 3, 7, 4, 6, 8, 1, 9.$$

The total weight is

$$2 + 5 + 3 + 7 + 4 + 6 + 8 + 1 + 9 = 45.$$

Step 4. A weighted median is reached once cumulative weight passes $45/2 = 22.5$. The cumulative weights are

$$2, 7, 10, 17, 21, 27, 35, 36, 45.$$

Thus the weighted median is

$$\frac{17}{6}.$$

Step 5. It is unique because the cumulative weight strictly crosses one-half at this value: the cumulative weight immediately before it is $21 < 22.5$ and immediately after it is $27 > 22.5$.

Final conclusion:

$$\boxed{\hat{\beta} = \frac{17}{6}},$$

and the minimizer is unique.

Part (b)

Theory required for this solution. For positive x_i,

$$\Delta(\beta) = \sum_{i=1}^{n} x_i \left| \frac{y_i}{x_i} - \beta \right|.$$

A weighted median minimizes this expression. If one weight is strictly larger than the sum of all other weights, the corresponding point is the unique weighted median, regardless of the values of the ratios.

Step 1. Here $x_i = 2^{i-1}$, so

$$x_n = 2^{n-1}.$$

The sum of all preceding weights is

$$\sum_{i=1}^{n-1} x_i = \sum_{i=0}^{n-2} 2^i = 2^{n-1} - 1.$$

Therefore

$$x_n = 2^{n-1} > 2^{n-1} - 1 = \sum_{i=1}^{n-1} x_i.$$

Step 2. Put

$$z_i = \frac{y_i}{x_i}.$$

Then

$$\Delta(\beta) = \sum_{i=1}^{n} x_i |z_i - \beta|.$$

The single weight x_n is larger than the combined weight of every other ratio.

Step 3. Consequently, the weighted median must be z_n. A direct way to see why is this: if $\beta < z_n$, increasing β toward z_n reduces the term $x_n|z_n - \beta|$ at rate x_n, whereas the total possible increase in the other terms is at most $\sum_{i<n} x_i < x_n$. The net change is therefore negative. Similarly, if $\beta > z_n$, moving down toward z_n decreases the objective.

Step 4. Since $z_n = y_n/x_n$,

$$\boxed{\hat{\beta} = \frac{y_n}{x_n}}.$$

Final conclusion: The minimizer is

$$\boxed{\arg\min_{\beta} \Delta(\beta) = \frac{y_n}{x_n}}.$$

The strict inequality $x_n > \sum_{i<n} x_i$ also gives uniqueness.

Part (c)

Theory required for this solution. Two facts are needed.

First, for the quadratic part,

$$\frac{d}{d\theta} \sum_{i=1}^{10} 2(x_i - \theta)^2 = 20(\theta - \bar{x}).$$

Second, for the absolute-value part, away from observations equal to θ,

$$\frac{d}{d\theta}\sum_{i=1}^{10}|x_i-\theta| = \#\{i : x_i < \theta\} - \#\{i : x_i > \theta\}.$$

For an even sample with median 3, at most five observations can lie below any $\theta < 3$, and at least five can lie at or below any $\theta > 3$. The objective is convex because it is a sum of convex functions.

Step 1. Suppose $\theta < 2$. Since $\bar{x} = 2$,

$$20(\theta - \bar{x}) = 20(\theta - 2) < 0.$$

Step 2. Because the median is 3, there cannot be six observations strictly below 2; otherwise the fifth and sixth order statistics would both be below 2, forcing the median below 2. Hence for $\theta < 2$,

$$\#\{x_i < \theta\} \leq 5,$$

so the absolute-value derivative/subgradient is non-positive.

Step 3. Therefore every subgradient of ψ at a point $\theta < 2$ is strictly negative: the quadratic contribution is already strictly negative, and the absolute-value contribution cannot offset it. Hence ψ is decreasing throughout $(-\infty, 2)$.

Why this step leads to the next step: If the function is decreasing as we move rightward up to 2, no point strictly below 2 can be a minimizer.

Step 4. Now suppose $\theta > 3$. The median condition implies at least five observations are at or below 3, so at a point strictly above 3 there cannot be more observations above θ than below it. Therefore the absolute-value contribution is non-negative. At the same time,

$$20(\theta - 2) > 0.$$

Thus every subgradient is strictly positive and ψ is increasing on $(3, \infty)$.

Step 5. Hence no minimizer can occur below 2 or above 3.

Final conclusion:

$$\boxed{\text{Every minimizer of } \psi \text{ lies in } [2, 3].}$$

Question 3

Whole question from the assignment.

Suppose that the correlation matrix of a random vector (X, Y, Z) is given by

$$\Sigma_0 = \begin{pmatrix} 1 & \rho & \rho \\ \rho & 1 & \rho \\ \rho & \rho & 1 \end{pmatrix}.$$

(a) Show that Σ_0 is a positive semi-definite matrix. (b) Show that ρ cannot be smaller than -0.5. (c) Assume that $\rho = -0.5$. Give a suitable choice of $a = (a_1, a_2, a_3)$ $(a \neq 0)$ such that variance of $a_1X + a_2Y + a_3Z$ is minimum.

Theory required for this solution. A covariance/correlation matrix is positive semi-definite, so for every vector a,

$$a^\top \Sigma_0 a \geq 0.$$

For the equicorrelation matrix

$$\Sigma_0 = (1-\rho)I + \rho \mathbf{1}\mathbf{1}^\top, \quad \mathbf{1} = (1,1,1)^\top,$$

the eigenvalues are $1-\rho$ (multiplicity 2) and $1+2\rho$ (multiplicity 1).

Part (a)

Step 1. For any $a = (a_1, a_2, a_3)^\top$,

$$a^\top \Sigma_0 a = (1-\rho)(a_1^2 + a_2^2 + a_3^2) + \rho(a_1 + a_2 + a_3)^2.$$

Step 2. Equivalently, decompose a into a component parallel to $\mathbf{1}$ and one orthogonal to $\mathbf{1}$. On the orthogonal subspace the eigenvalue is $1-\rho$; on the $\mathbf{1}$ direction the eigenvalue is $1+2\rho$.

Step 3. Since Σ_0 is a correlation matrix, it must be PSD, so its eigenvalues must be non-negative:

$$1-\rho \geq 0, \qquad 1+2\rho \geq 0.$$

Thus

$$-\frac{1}{2} \leq \rho \leq 1.$$

Final conclusion: $\boxed{\Sigma_0 \succeq 0}$, with the admissible equicorrelation range $\boxed{-1/2 \leq \rho \leq 1}$.

Part (b)

Step 1. Positive semidefiniteness requires

$$1 + 2\rho \geq 0.$$

Step 2. Therefore

$$2\rho \geq -1 \quad \Longrightarrow \quad \rho \geq -\frac{1}{2} = -0.5.$$

Final conclusion:

$$\boxed{\rho \geq -0.5}.$$

Part (c)

Step 1. If $\rho = -1/2$, the eigenvalue in the $\mathbf{1} = (1,1,1)^\top$ direction is

$$1 + 2\rho = 1 - 1 = 0.$$

Step 2. Therefore choose any non-zero vector proportional to $\mathbf{1}$:

$$a = (1,1,1)^o p.$$

Step 3. The variance is

$$\mathrm{Var}(a_1 X + a_2 Y + a_3 Z) = a^\top \Sigma_0 a.$$

Because a lies in the zero-eigenvalue direction,

$$a^\top \Sigma_0 a = 0.$$

Final conclusion:

$$\boxed{a = (1,1,1)^o p}$$

and the minimum variance is $\boxed{0}$.

Question 4

Whole question from the assignment.

(a) Show that in a simple linear regression problem, the product-moment correlation coefficient between the observed and estimated values of the response can never be negative. Also, show that the variance of the estimated values can never exceed the variance of the observed values.

(b) Consider a regression problem where the data set contains 100 bivariate observations, but X takes only two distinct values 1 and 2. Which of the two models, (i) $Y = \beta_0 + \beta_1 X$ or (ii) $Y = \alpha_0 X + \alpha_1 X^2$, will give a better fit (in terms of residual sum of squares) in this case? Justify your answer.

Part (a)

Theory required for this solution. In least-squares regression, the fitted vector is the orthogonal projection of Y onto the column space of the design matrix. Hence

$$Y = \hat{Y} + e, \qquad \hat{Y}^\top e = 0.$$

Therefore

$$Y^\top \hat{Y} = \left\|\hat{Y}\right\|^2 \geq 0,$$

and

$$\|Y\|^2 = \left\|\hat{Y}\right\|^2 + \|e\|^2 .$$

For a model with an intercept, the corresponding centred sum-of-squares decomposition gives

$$SST = SSR + SSE,$$

so $\mathrm{Var}(\hat{Y}) \leq \mathrm{Var}(Y)$.

Step 1. In simple regression with an intercept, the residual vector is orthogonal to the fitted vector:

$$\hat{Y}^\top e = 0, \qquad e = Y - \hat{Y}.$$

Thus

$$\hat{Y}^\top (Y - \hat{Y}) = 0 \quad \Longrightarrow \quad Y^\top \hat{Y} = \hat{Y}^\top \hat{Y}.$$

Step 2. Therefore the covariance/numerator in the correlation between the centred observed and fitted responses is non-negative. In fact, unless $\hat{Y}$ is constant, it is positive:

$$\mathrm{Corr}(Y, \hat{Y}) \geq 0.$$

Why this step leads to the next step: The fitted vector is a projection of Y. A vector and its orthogonal projection cannot have a negative inner product; the projection points in the same general direction as the original vector.

Step 3. Pythagoras gives

$$\|Y\|^2 = \left\|\hat{Y}\right\|^2 + \|e\|^2 \geq \left\|\hat{Y}\right\|^2.$$

After centring (because the regression contains an intercept), this is exactly

$$SST \geq SSR.$$

Dividing by the same sample-size factor yields

$$\mathrm{Var}(Y) \geq \mathrm{Var}(\hat{Y}).$$

Final conclusion:

$$\boxed{\mathrm{Corr}(Y, \hat{Y}) \geq 0}, \qquad \boxed{\mathrm{Var}(\hat{Y}) \leq \mathrm{Var}(Y)}.$$

Part (b)

Theory required for this solution. The least-squares fit depends only on the column space of the design matrix. If two design matrices have the same column space, they produce exactly the same projection matrix and therefore the same fitted values and RSS.

Step 1. Under model (i), the design columns are

$$\mathbf{1}, \quad X.$$

Since X takes only the values 1 and 2, this model can fit an arbitrary mean at the two observed X-levels.

Step 2. Under model (ii), the design columns are

$$X, \quad X^2.$$

At $X = 1$ they are $(1, 1)$ and at $X = 2$ they are $(2, 4)$. These two column patterns are linearly independent across the two levels, so the two-parameter model can again prescribe an arbitrary fitted value at $X = 1$ and another arbitrary fitted value at $X = 2$.

Indeed, if the desired fitted values are m_1 at $X = 1$ and m_2 at $X = 2$, solve

$$\alpha_0 + \alpha_1 = m_1, \qquad 2\alpha_0 + 4\alpha_1 = m_2.$$

This system has determinant $2 \neq 0$.

Step 3. Hence both models have the same fitted values: for each X-level they fit the corresponding sample mean. Therefore their RSS values are identical.

Final conclusion:

$$\boxed{\text{Neither model is better; the two models give exactly the same RSS.}}$$

Question 5

Whole question from the assignment.

Show that in the case of multiple linear regression, $\hat{Y}$, the vector of estimated values of the response, can be expressed as $\hat{Y} = P_X Y$, where Y is the vector of observed values of the response, and P_X is a matrix (known as the projection matrix) that satisfies:

(a) $P_X^2 = P_X$, $(I - P_X)^2 = I - P_X$ and $(I - P_X)P_X = 0$.

(b) Rank of P_X is the same as the rank of the data matrix X.

(c) $\text{Rank}(P_X) + \text{Rank}(I - P_X) = n$.

(d) $\hat{Y}^\top \hat{Y} = Y^\top P_X Y$ and $e^\top e = (Y - \hat{Y})^\top (Y - \hat{Y}) = Y^\top (I - P_X) Y$.

(e) For any vector v lying in the column space of X, $P_X v = v$ and $(I - P_X)v = 0$.

(f) If Y belongs to the column space of the data matrix X, then $\hat{Y} = Y$.

(g) If Y is regressed on $\hat{Y}$, the least square fit will be the $45°$ line passing through the origin.

Theory required for this solution. For full column rank X,

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y, \qquad \hat{Y} = X\hat{\beta} = P_X Y,$$

where

$$P_X = X(X^\top X)^{-1} X^\top.$$

This matrix is the orthogonal projection onto $\mathcal{C}(X)$, the column space of X. In particular $P_X^\top = P_X$, $P_X^2 = P_X$, and $I - P_X$ projects onto the orthogonal complement of $\mathcal{C}(X)$.

Step 1. Start from the least-squares estimator:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Multiplying by X,

$$\hat{Y} = X\hat{\beta} = X(X^\top X)^{-1} X^\top Y.$$

Define

$$\boxed{P_X = X(X^\top X)^{-1} X^\top}.$$

Hence

$$\boxed{\hat{Y} = P_X Y}.$$

Part (a)

Step 1. Compute P_X^2:

$$P_X^2 = X(X^\top X)^{-1} X^\top X (X^\top X)^{-1} X^\top = P_X.$$

Step 2. Now

$$(I - P_X)^2 = I - 2P_X + P_X^2 = I - P_X.$$

Step 3. Finally,

$$(I - P_X)P_X = P_X - P_X^2 = 0.$$

Final conclusion: All three identities follow directly from idempotence of P_X.

Part (b)

Step 1. Since $P_X = XA$ with $A = (X^\top X)^{-1}X^\top$, clearly

$$\mathcal{C}(P_X) \subseteq \mathcal{C}(X),$$

so $\operatorname{rank}(P_X) \le \operatorname{rank}(X)$.

Step 2. Conversely, if $v \in \mathcal{C}(X)$ then part (e) (proved below) gives $P_X v = v$. Thus every vector in $\mathcal{C}(X)$ is also in $\mathcal{C}(P_X)$, hence

$$\mathcal{C}(P_X) = \mathcal{C}(X).$$

Therefore

$$\boxed{\operatorname{rank}(P_X) = \operatorname{rank}(X)}.$$

Part (c)

Step 1. Since P_X and $I - P_X$ project onto complementary subspaces,

$$\mathcal{R}^n = \mathcal{C}(X) \oplus \mathcal{C}(X)^\perp.$$

Step 2. The dimensions therefore add:

$$\operatorname{rank}(P_X) + \operatorname{rank}(I - P_X) = n.$$

Final conclusion:

$$\boxed{\operatorname{rank}(P_X) + \operatorname{rank}(I - P_X) = n.}$$

Part (d)

Step 1. Because $P_X^\top = P_X$ and $P_X^2 = P_X$,

$$\hat{Y}^\top \hat{Y} = (P_X Y)^\top (P_X Y) = Y^\top P_X^\top P_X Y = Y^\top P_X Y.$$

Step 2. Since

$$e = Y - \hat{Y} = (I - P_X)Y,$$

we obtain

$$e^\top e = Y^\top (I - P_X)^\top (I - P_X) Y = Y^\top (I - P_X) Y.$$

Part (e)

Let $v \in \mathcal{C}(X)$. Then $v = Xc$ for some c. Hence

$$P_X v = X(X^\top X)^{-1} X^\top X c = Xc = v.$$

Therefore

$$(I - P_X)v = v - v = 0.$$

Part (f)

If $Y \in \mathcal{C}(X)$, part (e) gives

$$P_X Y = Y.$$

But $\hat{Y} = P_X Y$, so

$$\boxed{\hat{Y} = Y}.$$

Part (g)

Regress Y on $\hat{Y}$ through the origin. The least-squares slope is

$$\hat{b} = \frac{\hat{Y}^\top Y}{\hat{Y}^\top \hat{Y}}.$$

By part (d),

$$\hat{Y}^\top Y = \hat{Y}^\top \hat{Y}.$$

Hence

$$\hat{b} = 1.$$

Therefore the fitted line is

$$Y = \hat{Y},$$

which is the 45° line through the origin.

Final conclusion:

$$\boxed{Y = \hat{Y}}.$$

Question 6

Whole question from the assignment.

In the case of multiple linear regression, if $\hat{\beta}$ denotes the least squares estimate of β, show that for any other β_0, we have

$$\|Y - X\beta_0\|^2 = \left\|Y - X\hat{\beta}\right\|^2 + (\beta_0 - \hat{\beta})^\top (X^\top X)(\beta_0 - \hat{\beta}).$$

Theory required for this solution. The normal equations for least squares are

$$X^\top (Y - X\hat{\beta}) = 0.$$

Equivalently, the residual $e = Y - X\hat{\beta}$ is orthogonal to every column of X and hence to $X(\beta_0 - \hat{\beta})$ for every β_0.

Step 1. Write

$$Y - X\beta_0 = Y - X\hat{\beta} - X(\beta_0 - \hat{\beta}).$$

Let

$$e = Y - X\hat{\beta}, \qquad d = \beta_0 - \hat{\beta}.$$

Then

$$Y - X\beta_0 = e - Xd.$$

Step 2. Expand the squared norm:

$$\begin{aligned} \|Y - X\beta_0\|^2 &= \|e - Xd\|^2 \\ &= e^\top e - 2e^\top Xd + d^\top X^\top Xd. \end{aligned}$$

Step 3. From the normal equations,

$$e^\top X = (Y - X\hat{\beta})^\top X = 0^\top.$$

So the cross-term vanishes:

$$-2e^\top Xd = 0.$$

Step 4. Therefore

$$\|Y - X\beta_0\|^2 = e^\top e + d^\top X^\top Xd.$$

Substitute back e and d:

$$\boxed{\|Y - X\beta_0\|^2 = \left\|Y - X\hat{\beta}\right\|^2 + (\beta_0 - \hat{\beta})^\top X^\top X(\beta_0 - \hat{\beta}).}$$

Why this step leads to the next step: The second term is always non-negative because $X^\top X$ is positive semidefinite. Hence least squares is globally minimizing: every other parameter value adds a non-negative quadratic penalty to the minimum RSS.

Question 7

Whole question from the assignment.

(a) Show that in the case of simple linear regression (regression of Y on X), the solution of the normal equations always exists unless all observed values of X are equal.

(b) In a simple linear regression problem, can the two regression lines be (i) identical (ii) $2x + 3y = 7$ and $4x - 6y = 2$, (iii) $2x + 3y = 7$ and $4x + 6y = 15$? Give justification to your answer.

Part (a)

Theory required for this solution. For simple regression with intercept, the normal equations involve

$$X^\top X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}.$$

The determinant is

$$n\sum x_i^2 - (\sum x_i)^2 = n\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

This is positive exactly when the observed X values are not all equal.

Step 1. Let $\mathbf{1}$ be the n-vector of ones and let

$$X = \begin{pmatrix} \mathbf{1} & x \end{pmatrix}.$$

Then

$$X^\top X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}.$$

Step 2. Its determinant is

$$\det(X^\top X) = n\sum x_i^2 - (\sum x_i)^2.$$

Use

$$\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}.$$

Multiplying by n,

$$\det(X^\top X) = n\sum(x_i - \bar{x})^2.$$

Step 3. If not all x_i are equal, then $\sum(x_i - \bar{x})^2 > 0$, hence the determinant is positive and $X^\top X$ is invertible. Therefore the normal equations have a unique solution.

Step 4. If all x_i are equal, the intercept column and the X column are linearly dependent, so $X^\top X$ is singular.

Final conclusion: The normal equations have a solution (indeed a unique one under the usual full-rank setup) whenever the X observations are not all equal.

Part (b)

Theory required for this solution. The two regression lines must both pass through $(\bar{x}, \bar{y})$, and their slope coefficients satisfy

$$b_{Y|X} b_{X|Y} = r^2 \in [0, 1].$$

Thus the product of the two regression coefficients must be non-negative and cannot exceed 1. If the two lines are identical, $|r| = 1$.

Step 1. (i) Identical. Yes. Take any perfectly correlated sample with $r = 1$ or $r = -1$. Then the two regression lines coincide with the same exact line.

Step 2. (ii) Rewrite the two equations as regression forms:

$$2x + 3y = 7 \iff y = \frac{7}{3} - \frac{2}{3}x,$$

so $b_{Y|X} = -2/3$. Also,

$$4x - 6y = 2 \iff x = \frac{1}{2} + \frac{3}{2}y,$$

so $b_{X|Y} = 3/2$. Their product is

$$\left(-\frac{2}{3}\right)\left(\frac{3}{2}\right) = -1,$$

which cannot equal $r^2 \geq 0$. Therefore impossible.

Step 3. (iii) Now

$$2x + 3y = 7 \iff y = \frac{7}{3} - \frac{2}{3}x,$$

and

$$4x + 6y = 15 \iff x = \frac{15}{4} - \frac{3}{2}y.$$

The product of the slope coefficients is

$$\left(-\frac{2}{3}\right)\left(-\frac{3}{2}\right) = 1,$$

which would require $|r| = 1$, so the two regression lines would have to coincide. But the two displayed lines are parallel and distinct because

$$4x + 6y = 15 \iff 2x + 3y = \frac{15}{2} \neq 7.$$

This contradicts the fact that both regression lines must pass through the same mean point.

Final conclusion:

$$\boxed{\text{(i) Yes;} \quad \text{(ii) No;} \quad \text{(iii) No.}}$$

Question 8

Whole question from the assignment.

Consider a regression problem based on observations on three variables X_1, X_2, Y. Assume that the observations on X_1, X_2, Y are centered so that you can ignore the intercept term while fitting the regression model. Also assume that $\mathrm{Corr}(X_1, X_2)$, $\mathrm{Corr}(X_1, Y)$ and $\mathrm{Corr}(X_2, Y)$ are all positive.

(a) Show that if you regress Y on X_1, the regression coefficient β_{yx_1} turns out to be positive. Similarly, one can show that if you regress Y on X_2, the corresponding coefficient β_{yx_2} turns out to be positive as well.

(b) If you regress Y on X_1 and X_2 simultaneously (multiple linear regression), can any of the estimated regression coefficients be negative? Justify your answer.

(c) Let the residual sum of squares in the first two regressions be RSS_1 and RSS_2, respectively and that in the case of multiple linear regression be RSS_{12}. Show that $RSS_{12} \leq \min\{RSS_1, RSS_2\}$.

Part (a)

Theory required for this solution. With centered variables and no intercept, the simple-regression slope is

$$\hat{\beta}_{Y|X} = \frac{\sum X_i Y_i}{\sum X_i^2} = r_{XY}\frac{s_Y}{s_X}.$$

Since standard deviations are positive, the sign of the slope equals the sign of the correlation.

Step 1. Since $\mathrm{Corr}(X_1, Y) > 0$,

$$\hat{\beta}_{yx_1} = \mathrm{Corr}(X_1, Y)\frac{s_Y}{s_{X_1}} > 0.$$

Similarly,

$$\hat{\beta}_{yx_2} = \mathrm{Corr}(X_2, Y)\frac{s_Y}{s_{X_2}} > 0.$$

Final conclusion:

$$\boxed{\hat{\beta}_{yx_1} > 0, \qquad \hat{\beta}_{yx_2} > 0.}$$

Part (b)

Theory required for this solution. For standardized centered predictors,

$$\hat{\beta}_1 = \frac{r_1 - r r_2}{1 - r^2}, \qquad \hat{\beta}_2 = \frac{r_2 - r r_1}{1 - r^2},$$

where $r = \mathrm{Corr}(X_1, X_2)$, $r_1 = \mathrm{Corr}(X_1, Y)$ and $r_2 = \mathrm{Corr}(X_2, Y)$. The signs of multiple-regression coefficients need not match the signs of marginal correlations because each coefficient measures a partial effect after adjusting for the other predictor.

Step 1. Yes, a coefficient can be negative even though all three pairwise correlations are positive.

Step 2. A concrete valid example is

$$r = 0.8, \qquad r_1 = 0.7, \qquad r_2 = 0.9.$$

Then

$$\hat\beta_1 = \frac{0.7-(0.8)(0.9)}{1-0.8^2} = \frac{-0.02}{0.36} < 0,$$

while

$$\hat\beta_2 = \frac{0.9-(0.8)(0.7)}{0.36} > 0.$$

Step 3. The 3×3 correlation matrix for this example is positive definite because its determinant is

$$1 + 2(0.8)(0.7)(0.9) - 0.8^2 - 0.7^2 - 0.9^2 = 0.068 > 0.$$

So the example is not algebraically impossible.

Final conclusion:

$\boxed{\text{Yes. Multiple-regression coefficients can be negative despite positive pairwise correlations.}}$

Part (c)

Theory required for this solution. Least squares minimizes RSS over the chosen column space. If one model's design matrix has a column space contained in another's, the larger model cannot have larger minimum RSS.

Step 1. The model using only X_1 is a submodel of the model using both X_1 and X_2. Therefore the set of candidate fitted vectors for the one-predictor model is contained in the set for the two-predictor model.

Step 2. Hence the minimum residual sum of squares over the larger set is no bigger:

$$RSS_{12} \le RSS_1.$$

Similarly,

$$RSS_{12} \le RSS_2.$$

Step 3. Combining,

$$\boxed{RSS_{12} \le \min\{RSS_1, RSS_2\}.}$$

Question 9

Whole question from the assignment.

Consider two sets of independent observations on (X, Y). Let $\hat\beta_1$ be the least-squares estimate of the regression coefficient β computed from the first set and $\hat\beta_2$ be that computed from the second data set.

(a) If $\hat\beta_1$ and $\hat\beta_2$ both are positive, can the estimate of the regression coefficient computed from the pooled data be negative? Justify your answer.

(b) If the means of X observations are equal in these data sets, will you modify your answer?

Theory required for this solution. For a pooled simple regression with intercept,

$$S_{XY}^{\text{pool}} = S_{XY,1} + S_{XY,2} + \frac{n_1 n_2}{n_1+n_2}(\bar X_1 - \bar X_2)(\bar Y_1 - \bar Y_2).$$

Also

$$\hat\beta_j = \frac{S_{XY,j}}{S_{XX,j}}.$$

Thus positive within-group slopes make the within-group cross-products positive, but the between-group term can be negative and can dominate.

Part (a)

Step 1. Since $\hat{\beta}_j > 0$,

$$S_{XY,j} = \hat{\beta}_j S_{XX,j} > 0, \qquad j = 1, 2.$$

Step 2. But pooling adds the between-group term

$$B = \frac{n_1 n_2}{n_1 + n_2}(\bar{X}_1 - \bar{X}_2)(\bar{Y}_1 - \bar{Y}_2).$$

This can be negative.

Step 3. It can be so negative that

$$S_{XY,1} + S_{XY,2} + B < 0,$$

which makes the pooled slope negative.

Step 4. Example: take two groups of size 2. Group 1:

$$(9, -11), \quad (11, -9),$$

so the slope within this group is $+1$. Group 2:

$$(-1, -1), \quad (1, 1),$$

so its slope is also $+1$. Yet the group means are $(10, -10)$ and $(0, 0)$, generating a large negative between-group cross-product. The pooled values are

$$S_{XY,1} = 2, \quad S_{XY,2} = 2, \quad B = (1)(10)(-10) = -100,$$

so

$$S_{XY}^{\text{pool}} = -96 < 0.$$

The pooled slope is therefore negative.

Final conclusion:

$$\boxed{\text{Yes, the pooled slope can be negative.}}$$

This is a standard between-group effect / Simpson-type phenomenon.

Part (b)

Step 1. If $\bar{X}_1 = \bar{X}_2$, then the between-group term is exactly zero:

$$B = 0.$$

Step 2. Hence

$$S_{XY}^{\text{pool}} = S_{XY,1} + S_{XY,2} > 0.$$

Therefore the pooled slope cannot be negative.

Final conclusion:

$$\boxed{\text{Yes, the answer changes: with equal } X \text{ means the pooled slope is positive (unless degenerate).}}$$

Question 10

Whole question from the assignment.

Consider a linear regression model $Y_{n\times1} = X_{n\times2}(\beta_1, \beta_2)^\top + \epsilon_{n\times1}$ with two non-stochastic covariates and no intercept, where $E(\epsilon) = 0$ and $\mathrm{Disp}(\epsilon) = \sigma^2 I$. Assume that the values of all three variables X_1, X_2, Y are standardised (i.e. they have sample mean 0 and sample variance 1). Let r_i be the sample correlation between X_i and Y $(i = 1, 2)$ and r be the sample correlation between X_1 and X_2. If $\hat\beta_1 > \hat\beta_2 > 0$ are the β_1 and β_2, respectively, comment on the validity of:

(a) r_1 and r_2 are positive. (b) r_1 is larger than r_2. (c) The ridge regression estimates of β_1 and β_2 are positive. (d) For any $\ell = (\ell_1, \ell_2)$ with $\|\ell\| = 1$, the variance of $\ell_1\hat\beta_1 + \ell_2\hat\beta_2$ cannot be smaller than $\sigma^2/[n(1 + |r|)]$.

Theory required for this solution. Because the variables are standardised,

$$\frac{1}{n}X^\top X = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \frac{1}{n}X^\top Y = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}.$$

Hence

$$\hat\beta = \frac{1}{1-r^2}\begin{pmatrix} r_1 - rr_2 \\ r_2 - rr_1 \end{pmatrix}.$$

Equivalently,

$$r_1 = \hat\beta_1 + r\hat\beta_2, \qquad r_2 = r\hat\beta_1 + \hat\beta_2.$$

For ridge, $X^\top X$ is replaced by $X^\top X + \lambda I$. For the covariance of OLS,

$$\mathrm{Var}(\hat\beta) = \sigma^2(X^\top X)^{-1} = \frac{\sigma^2}{n(1-r^2)}\begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}.$$

Part (a)

Step 1. From

$$r_2 = r\hat\beta_1 + \hat\beta_2,$$

if $r < 0$ with sufficiently large magnitude, $r\hat\beta_1$ can outweigh $\hat\beta_2$. For example,

$$r = -0.9, \quad \hat\beta_1 = 0.2, \quad \hat\beta_2 = 0.1$$

gives

$$r_2 = -0.18 + 0.1 = -0.08 < 0.$$

So r_2 need not be positive.

Final conclusion:

$$\boxed{\text{(a) False.}}$$

Part (b)

Step 1. Subtract the two equations:

$$r_1 - r_2 = (\hat\beta_1 - \hat\beta_2)(1 - r).$$

Step 2. Since $\hat\beta_1 > \hat\beta_2$ and a valid correlation has $r < 1$,

$$(\hat\beta_1 - \hat\beta_2)(1 - r) > 0.$$

Hence

$$r_1 > r_2.$$

Final conclusion:

$$\boxed{\text{(b) True.}}$$

Part (c)

Step 1. Ridge solves

$$\hat{\beta}_R = (X^\top X + \lambda I)^{-1} X^\top Y.$$

Put $k = \lambda/n > 0$. Then

$$\hat{\beta}_R = \left[\begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} + kI\right]^{-1} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}.$$

Step 2. To test the claim, write the ridge estimator explicitly. Since

$$(X^\top X + \lambda I)/n = \begin{pmatrix} 1+k & r \\ r & 1+k \end{pmatrix}, \qquad k = \frac{\lambda}{n} > 0,$$

we obtain

$$\hat{\beta}_R = \frac{1}{(1+k)^2 - r^2} \begin{pmatrix} (1+k)r_1 - rr_2 \\ (1+k)r_2 - rr_1 \end{pmatrix}.$$

Step 3. Substitute the OLS identities

$$r_1 = \hat{\beta}_1 + r\hat{\beta}_2, \qquad r_2 = r\hat{\beta}_1 + \hat{\beta}_2.$$

The second ridge numerator becomes

$$\begin{aligned}(1+k)r_2 - rr_1 &= (1+k)(r\hat{\beta}_1 + \hat{\beta}_2) - r(\hat{\beta}_1 + r\hat{\beta}_2) \\ &= kr\hat{\beta}_1 + (1 + k - r^2)\hat{\beta}_2.\end{aligned}$$

Thus, when $r < 0$, the first term is negative and can dominate the second.

Step 4. Take

$$r = -0.9, \qquad \hat{\beta}_1 = 0.2, \qquad \hat{\beta}_2 = 0.1, \qquad k = 1.$$

Then

$$(1+k)r_2 - rr_1 = (1 + 1 - 0.81)(0.1) + (1)(-0.9)(0.2) = 0.119 - 0.18 < 0.$$

The denominator $(1+k)^2 - r^2$ is positive, so the second ridge coefficient is negative. Hence the assertion that both ridge estimates must be positive is false.

Final conclusion:

$$\boxed{\text{(c) False. Ridge coefficients need not both remain positive.}}$$

Part (d)

Step 1. For any unit vector ℓ,

$$\operatorname{Var}(\ell^\top \hat{\beta}) = \ell^\top \operatorname{Var}(\hat{\beta})\ell.$$

Step 2. The eigenvalues of

$$\frac{1}{n(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}$$

are

$$\frac{1}{n(1+r)}, \qquad \frac{1}{n(1-r)}.$$

The smaller of these is

$$\frac{1}{n(1+|r|)}.$$

Therefore the Rayleigh quotient inequality gives

$$\operatorname{Var}(\ell^\top \hat{\beta}) \geq \frac{\sigma^2}{n(1+|r|)} \qquad \text{for all } \|\ell\| = 1.$$

Final conclusion:

$$\boxed{\text{(d) True.}}$$

Equality occurs for the unit eigenvector associated with the smaller eigenvalue.

Question 11

Whole question from the assignment.

(a) Consider a multiple linear regression problem, where the design matrix X is such that $X^\top X = I$, the identity matrix. Show that the LASSO estimates of the parameters of this multiple linear regression model can be obtained by using a smoothing operator S_λ on the coordinates of the usual least squares estimate, where

$$S_\lambda(t) = \begin{cases} t - \lambda/2 \ \operatorname{sign}(t), & |t| > \lambda/2, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Discuss how the coordinate descent method can be used for computing the LASSO estimates of the model parameters when $X^\top X \neq I$.

Theory required for this solution. The course notes define LASSO through

$$\min_\beta \ \|Y - X\beta\|^2 + \lambda \|\beta\|_1 .$$

When $X^\top X = I$, the least-squares estimate is $\hat{\beta}_{LS} = X^\top Y$ and the objective separates coordinatewise. The one-dimensional minimizer of

$$(u-t)^2 + \lambda |u|$$

is soft thresholding:

$$S_\lambda(t) = \operatorname{sign}(t) \max\{|t| - \lambda/2, 0\}.$$

Part (a)

Step 1. Let $\hat{\beta}_{LS} = X^\top Y$. Expand:

$$\|Y - X\beta\|^2 = Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X\beta.$$

Because $X^\top X = I$,

$$\|Y - X\beta\|^2 = \text{constant} + \left\|\beta - \hat{\beta}_{LS}\right\|^2.$$

Step 2. Therefore the LASSO problem is equivalent to

$$\min_\beta \sum_j \left[(\beta_j - \hat{\beta}_{LS,j})^2 + \lambda |\beta_j|\right],$$

which separates across coordinates.

Step 3. For one coordinate, minimize

$$f(u) = (u - t)^2 + \lambda|u|.$$

If $u > 0$,

$$f'(u) = 2(u - t) + \lambda,$$

so $u = t - \lambda/2$, provided this is positive. If $u < 0$,

$$f'(u) = 2(u - t) - \lambda,$$

so $u = t + \lambda/2$, provided this is negative. If $|t| \leq \lambda/2$, the minimum occurs at $u = 0$.

Step 4. Hence

$$\hat{\beta}_{L,j} = S_\lambda(\hat{\beta}_{LS,j}).$$

Final conclusion:

$$\boxed{\hat{\beta}_L = S_\lambda(\hat{\beta}_{LS}) \text{ coordinatewise.}}$$

Part (b)

Step 1. When $X^\top X \neq I$, coordinates are coupled through the quadratic term. Coordinate descent breaks the full optimization problem into one-dimensional minimizations.

Step 2. Fix all coefficients except β_j and define the partial residual

$$r_j = Y - \sum_{k \neq j} X_k \beta_k.$$

The coordinate subproblem is

$$\min_u \ \|r_j - X_j u\|^2 + \lambda|u|.$$

Step 3. Expanding,

$$\|r_j - X_j u\|^2 = \text{constant} - 2uX_j^\top r_j + u^2 \|X_j\|^2.$$

Thus the update is

$$\boxed{\beta_j \leftarrow \frac{1}{\|X_j\|^2} S_\lambda(X_j^\top r_j)}$$

under the scaling convention of the question. Equivalently,

$$\beta_j \leftarrow \frac{1}{\|X_j\|^2} \operatorname{sign}(z_j) \max\left\{|z_j| - \frac{\lambda}{2}, 0\right\}, \quad z_j = X_j^\top r_j,$$

with the threshold appropriately divided by $\|X_j\|^2$ if one defines the smoothing operator directly in terms of the unnormalised coordinate.

Step 4. Cycle through $j = 1, \ldots, p$ repeatedly until the objective or the parameter vector changes negligibly.

Final conclusion: Coordinate descent is an iterative sequence of one-dimensional soft-thresholding updates. It is used because LASSO has no general closed-form solution when the predictors are correlated.

Question 12

Whole question from the assignment.

Consider a linear model $Y_i = \beta_0+\beta_1 x_{1i}+\cdots+\beta_p x_{pi}+\epsilon_i$, $i = 1, 2, \ldots, n$, where the predictor variables are non-stochastic and ϵ_i's are iid G, for G being a univariate distribution with mean 0 and variance σ^2.

(a) If G is normal, show that the maximum likelihood estimate of $(\beta_0, \ldots, \beta_p)$ coincides with the least square estimate. (b) If G is double exponential, show that the maximum likelihood estimate of $(\beta_0, \ldots, \beta_p)$ coincides with the LAD estimate. (c) If G is normal, show that ridge and LASSO estimates of $(\beta_0, \ldots, \beta_p)$ can be viewed as the MAP estimates for suitable choices of the prior distributions.

Part (a)

Theory required for this solution. For normal errors,

$$\epsilon_i \sim N(0, \sigma^2),$$

so

$$f(y_i \mid x_i, \beta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - x_i^\top \beta)^2}{2\sigma^2}\right].$$

The likelihood is the product over i, so the log-likelihood differs from

$$-\frac{1}{2\sigma^2} \sum_i (y_i - x_i^\top \beta)^2$$

only by constants. Maximizing likelihood is therefore equivalent to minimizing RSS.

Step 1. Let $x_i^\top = (1, x_{1i}, \ldots, x_{pi})$ and $r_i = y_i - x_i^\top \beta$. Then

$$L(\beta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{r_i^2}{2\sigma^2}\right).$$

Step 2. Take logs:

$$\ell(\beta) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} r_i^2.$$

For fixed σ^2, maximizing $\ell(\beta)$ is equivalent to minimizing

$$\sum_{i=1}^{n} r_i^2 = \mathrm{RSS}(\beta).$$

Final conclusion:

$$\boxed{\hat\beta_{MLE} = \hat\beta_{LS}.}$$

Part (b)

Theory required for this solution. A double-exponential (Laplace) density with scale b is

$$f(\epsilon) = \frac{1}{2b} e^{-|\epsilon|/b}.$$

Thus the log-likelihood is constant minus a positive multiple of

$$\sum_i |y_i - x_i^\top \beta|.$$

Therefore MLE coincides with least absolute deviations (LAD).

Step 1. The likelihood is

$$L(\beta) = \prod_{i=1}^{n} \frac{1}{2b} \exp\left(-\frac{|y_i - x_i^\top \beta|}{b}\right).$$

Step 2. Taking logs,

$$\ell(\beta) = -n \log(2b) - \frac{1}{b} \sum_i |y_i - x_i^\top \beta|.$$

Therefore maximizing likelihood is exactly the same as minimizing the LAD criterion.

Final conclusion:

$$\boxed{\hat{\beta}_{MLE} = \hat{\beta}_{LAD}.}$$

Part (c)

Theory required for this solution. MAP estimation maximizes

$$p(\beta \mid Y) \propto L(\beta)\, \pi(\beta),$$

so it minimizes negative log-likelihood plus negative log-prior. Under a normal likelihood:

$$-\log L(\beta) = \frac{1}{2\sigma^2} \mathrm{RSS}(\beta) + \text{constant}.$$

A Gaussian prior produces an ℓ_2 penalty; a Laplace prior produces an ℓ_1 penalty.

Step 1. Ridge. Give the coefficients a Gaussian prior, for example

$$\beta_j \overset{ind}{\sim} N(0, \tau^2),$$

with a flat prior for the intercept if desired. Then

$$-\log \pi(\beta) = \frac{1}{2\tau^2} \sum_{j=1}^{p} \beta_j^2 + \text{constant}.$$

Therefore the negative log-posterior is proportional to

$$\mathrm{RSS}(\beta) + \frac{\sigma^2}{\tau^2} \sum_{j=1}^{p} \beta_j^2.$$

This is ridge with

$$\lambda = \frac{\sigma^2}{\tau^2}.$$

Step 2. LASSO. Give the coefficients independent Laplace priors,

$$\pi(\beta_j) \propto e^{-|\beta_j|/b_0}.$$

Then

$$-\log \pi(\beta) = \frac{1}{b_0} \sum_j |\beta_j| + \text{constant}.$$

The negative log-posterior is therefore proportional to

$$\text{RSS}(\beta) + \frac{2\sigma^2}{b_0} \sum_j |\beta_j|.$$

Thus LASSO is obtained with

$$\lambda = \frac{2\sigma^2}{b_0}.$$

Final conclusion:

$$\boxed{\text{Gaussian prior} \iff \text{ridge MAP}, \qquad \text{Laplace prior} \iff \text{LASSO MAP}.}$$

Question 13

Whole question from the assignment.

(a) The sub-gradient of a function $f : \mathbb{R} \to \mathbb{R}$ at a point x_0 is defined as the set

$$\partial f(x)|_{x=x_0} = \{c \in \mathbb{R} : f(x) - f(x_0) \geq c(x - x_0) \text{ for all } x \in \mathbb{R}\}.$$

Show that 0 belongs to the set if and only if x_0 is a minimiser of f. (b) Show that if f is a convex function and the derivative $f'(x_0)$ exists at x_0, then $\partial f(x)|_{x=x_0} = \{f'(x_0)\}$. (c) Consider two functions f_1 and f_2. Define $f = f_1 + f_2$. Show that if $c_1 \in \partial f_1(x)|_{x=x_0}$ and $c_2 \in \partial f_2(x)|_{x=x_0}$, then $c_1 + c_2 \in \partial f(x)|_{x=x_0}$.

Part (a)

Theory required for this solution. By definition, $0 \in \partial f(x_0)$ means

$$f(x) - f(x_0) \geq 0 \cdot (x - x_0) = 0 \quad \text{for all } x,$$

which is precisely the statement that $f(x_0) \leq f(x)$ for every x.

Step 1. $0 \in \partial f(x_0)$ iff

$$f(x) - f(x_0) \geq 0 \qquad \forall x.$$

Step 2. This is equivalent to

$$f(x_0) \leq f(x) \qquad \forall x,$$

which is exactly the definition of x_0 being a global minimizer.

Final conclusion:

$$\boxed{0 \in \partial f(x_0) \iff x_0 \text{ is a minimizer of } f.}$$

Part (b)

Theory required for this solution. For a differentiable convex function, the tangent line is a global supporting line:

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0).$$

Thus $f'(x_0)$ is a subgradient. To show uniqueness, use the definition and take one-sided difference quotients as $x \to x_0$.

Step 1. Convexity and differentiability imply

$$f(x) - f(x_0) \geq f'(x_0)(x - x_0),$$

so

$$f'(x_0) \in \partial f(x_0).$$

Step 2. Let $c \in \partial f(x_0)$. Then

$$f(x) - f(x_0) \geq c(x - x_0).$$

For $x > x_0$, divide by $x - x_0 > 0$:

$$\frac{f(x) - f(x_0)}{x - x_0} \geq c.$$

Let $x \downarrow x_0$ to get $f'(x_0) \geq c$.

Step 3. For $x < x_0$, dividing by the negative quantity $x - x_0$ reverses the inequality, giving

$$\frac{f(x) - f(x_0)}{x - x_0} \leq c.$$

Let $x \uparrow x_0$ to get $f'(x_0) \leq c$. Therefore $c = f'(x_0)$.

Final conclusion:

$$\boxed{\partial f(x_0) = \{f'(x_0)\}.}$$

Part (c)

Theory required for this solution. If $c_1 \in \partial f_1(x_0)$ and $c_2 \in \partial f_2(x_0)$, then

$$f_1(x) - f_1(x_0) \geq c_1(x - x_0),$$

$$f_2(x) - f_2(x_0) \geq c_2(x - x_0).$$

Adding inequalities is the key subgradient rule for sums.

Step 1. Add the two defining inequalities:

$$f_1(x) - f_1(x_0) + f_2(x) - f_2(x_0) \geq (c_1 + c_2)(x - x_0).$$

Step 2. Since $f = f_1 + f_2$,

$$f(x) - f(x_0) \geq (c_1 + c_2)(x - x_0).$$

Therefore

$$\boxed{c_1 + c_2 \in \partial f(x_0).}$$

Question 14

Whole question from the assignment.

The sub-gradient of a function $f : \mathbb{R}^p \to \mathbb{R}$ at a point x_0 is defined as the set

$$\partial f(x)|_{x=x_0} = \{c \in \mathbb{R}^p : f(x) - f(x_0) \geq c^\top (x - x_0) \text{ for all } x \in \mathbb{R}^p\}.$$

Show that $0 = (0, 0, \ldots, 0) \in \mathbb{R}^p$ belongs to the set if and only if x_0 is a minimiser of f.

Theory required for this solution. The scalar argument from Question 13 is unchanged in vector form. By definition,

$$0 \in \partial f(x_0) \iff f(x) - f(x_0) \geq 0^\top (x - x_0) = 0 \quad \forall x.$$

Step 1. Start from

$$0 \in \partial f(x_0).$$

This means

$$f(x) - f(x_0) \geq 0^\top (x - x_0) = 0 \qquad \forall x \in \mathbb{R}^p.$$

Step 2. Therefore

$$f(x) \geq f(x_0) \qquad \forall x,$$

which says precisely that x_0 is a global minimizer.

Step 3. Conversely, if x_0 is a minimizer, then $f(x) - f(x_0) \geq 0$ for all x, so the defining subgradient inequality holds with $c = 0$.

Final conclusion:

$$\boxed{0 \in \partial f(x_0) \iff x_0 \text{ is a global minimizer of } f.}$$

Question 15

Whole question from the assignment.

(a) Find the sub-gradient of the function $f(x) = \|x\|_1$ at x_0 when (i) $x_0 = 0$ and (ii) $x_0 \neq 0$, where $\|x\|_1$ denotes the ℓ_1-norm of x.

(b) Find the sub-gradient of the function $f(x) = \|x\|_2$ at x_0 when (i) $x_0 = 0$ and (ii) $x_0 \neq 0$, where $\|x\|_2$ denotes the ℓ_2-norm (i.e., the Euclidean norm of x).

Part (a)

Theory required for this solution.

$$\|x\|_1 = \sum_{j=1}^{p} |x_j|.$$

For the scalar absolute value,

$$\partial|t| = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

The subgradient of a sum is obtained coordinatewise.

Step 1. If $x_0 = 0$, every coordinate is at the kink of $|t|$. Therefore

$$\boxed{\partial \|x\|_1 \big|_{x=0} = [-1, 1]^p.}$$

Step 2. If $x_0 \neq 0$, each nonzero coordinate contributes its sign. Therefore

$$\boxed{\partial \|x\|_1 \big|_{x=x_0} = \left\{ g \in \mathbb{R}^p : \begin{array}{ll} g_j = \operatorname{sign}(x_{0j}), & x_{0j} \neq 0, \\ g_j \in [-1, 1], & x_{0j} = 0 \end{array} \right\}.}$$

Part (b)

Theory required for this solution. For $x \neq 0$,

$$\|x\|_2 = (x^\top x)^{1/2} \quad \Longrightarrow \quad \nabla \|x\|_2 = \frac{x}{\|x\|_2}.$$

At $x = 0$ the Euclidean norm is not differentiable. Its subdifferential is the closed unit Euclidean ball.

Step 1. For $x_0 \neq 0$,

$$\boxed{\partial \|x\|_2 \big|_{x=x_0} = \left\{ \frac{x_0}{\|x_0\|_2} \right\}.}$$

Step 2. At $x_0 = 0$, the subgradient condition is

$$\|x\|_2 \geq g^\top x \qquad \forall x.$$

By Cauchy–Schwarz,

$$g^\top x \leq \|g\|_2 \|x\|_2 .$$

The inequality holds for all x exactly when $\|g\|_2 \leq 1$. Hence

$$\boxed{\partial \|x\|_2 \big|_{x=0} = \{g \in \mathbb{R}^p : \|g\|_2 \leq 1\}.}$$

Question 16

Whole question from the assignment.

(a) Describe the coordinate descent algorithm for elastic net regression. (b) Give an example of a regression problem where the LASSO solution is not unique but the elastic net solution is unique.

Part (a)

Theory required for this solution. The course notes define elastic net by minimizing

$$\|Y - X\beta\|^2 + \lambda \left(\frac{\alpha}{2} \|\beta\|_2^2 + (1 - \alpha) \|\beta\|_1 \right), \quad \lambda > 0,$$

where α controls the mixture of ℓ_2 and ℓ_1 penalties. Coordinate descent updates one coefficient at a time while fixing the others.

Step 1. Start with an initial vector $\beta^{(0)}$, often zero.

Step 2. For coordinate j, form the partial residual

$$r_j = Y - \sum_{k \neq j} X_k \beta_k.$$

Then update β_j by minimizing

$$\|r_j - X_j\beta_j\|^2 + \lambda \left(\frac{\alpha}{2}\beta_j^2 + (1-\alpha)|\beta_j| \right).$$

Step 3. Differentiating the quadratic part and using the subgradient of $|\beta_j|$ gives the soft-threshold update

$$\boxed{\beta_j \leftarrow \frac{S_{\lambda(1-\alpha)}(X_j^\top r_j/\|X_j\|^2)}{1 + \lambda\alpha/(2\|X_j\|^2)}}$$

under a matching definition of the soft-threshold parameter. More transparently, with $z_j = X_j^\top r_j$,

$$\boxed{\beta_j \leftarrow \frac{\operatorname{sign}(z_j)\max\{|z_j| - \lambda(1-\alpha)/2, 0\}}{\|X_j\|^2 + \lambda\alpha/2}.}$$

Step 4. Cycle through all coordinates repeatedly until convergence.

> **Final conclusion:** The elastic-net coordinate update is simply a soft-thresholding numerator divided by an enlarged quadratic coefficient coming from the ℓ_2 penalty.

Part (b)

Step 1. Take two identical predictors:

$$X_1 = X_2 = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Step 2. The LASSO objective depends on $\beta_1 + \beta_2$ and $|\beta_1| + |\beta_2|$. For an appropriate range of λ (e.g. $0 < \lambda < 2n$ under this scaling), every nonnegative split satisfying

$$\beta_1 + \beta_2 = 1 - \frac{\lambda}{2n}$$

has the same LASSO objective. Hence the solution is not unique.

Step 3. Add the elastic-net ℓ_2 penalty. For a fixed sum $s = \beta_1 + \beta_2$,

$$\beta_1^2 + \beta_2^2$$

is uniquely minimized at

$$\beta_1 = \beta_2 = \frac{s}{2}.$$

Thus the elastic-net minimizer is unique.

> **Final conclusion:**
>
> $$\boxed{X_1 = X_2 \text{ is an example where LASSO is non-unique but elastic net is unique.}}$$
>
> The ℓ_2 part makes the criterion strictly convex in the coefficients (when $\alpha > 0$), breaking the flat LASSO solution set.

Question 17

Whole question from the assignment.

Consider a regression problem with two covariates, where $X^\top X = I$, the identity matrix. Suppose that all variables are centred, and the least squares fit of Y on X_1 and X_2 is given by $Y = 0.9X_1 - 0.6X_2$. Find the minimizer of:
(a) Minimize $\sum_{i=1}^{n}[Y_i - \beta_1 X_{1i} - \beta_2 X_{2i}]^2$ subject to $\beta_1^2 + \beta_2^2 = 1$.
(b) Minimize $\sum_{i=1}^{n}[Y_i - \beta_1 X_{1i} - \beta_2 X_{2i}]^2$ subject to $|\beta_1| + |\beta_2| = 1$.

Theory required for this solution. When $X^\top X = I$,

$$\|Y - X\beta\|^2 = \left\|Y - X\hat{\beta}_{LS}\right\|^2 + \left\|\beta - \hat{\beta}_{LS}\right\|^2.$$

Therefore constrained least squares reduces to finding the point on the constraint set closest in Euclidean distance to the OLS vector. Here

$$\hat{\beta}_{LS} = (0.9, -0.6)^\top.$$

Part (a)

Step 1. Minimize

$$\left\|\beta - (0.9, -0.6)^\top\right\|^2$$

subject to $\|\beta\|_2 = 1$.

Step 2. The closest point on the unit circle to a nonzero vector points in the same direction. Hence normalize the OLS vector:

$$\left\|\hat{\beta}_{LS}\right\|_2 = \sqrt{0.9^2 + (-0.6)^2} = \sqrt{1.17}.$$

Therefore

$$\boxed{\beta^* = \frac{(0.9, -0.6)}{\sqrt{1.17}} \approx (0.83205, -0.55470).}$$

Part (b)

Step 1. The OLS point $(0.9, -0.6)$ lies outside the ℓ_1 ball because

$$|0.9| + |-0.6| = 1.5 > 1.$$

Step 2. The nearest point must have the same signs, so take $\beta_1 \geq 0$ and $\beta_2 \leq 0$. The constraint becomes

$$\beta_1 - \beta_2 = 1 \quad \Longrightarrow \quad \beta_2 = \beta_1 - 1.$$

Step 3. Substitute into the squared distance:

$$f(\beta_1) = (\beta_1 - 0.9)^2 + (\beta_1 - 1 + 0.6)^2 = (\beta_1 - 0.9)^2 + (\beta_1 - 0.4)^2.$$

Differentiate:

$$f'(\beta_1) = 2(\beta_1 - 0.9) + 2(\beta_1 - 0.4) = 4\beta_1 - 2.6.$$

Set equal to zero:

$$\beta_1 = 0.65.$$

Then

$$\beta_2 = 0.65 - 1 = -0.35.$$

Final conclusion:

$$\boxed{\beta^* = (0.65, -0.35).}$$

Question 18

Whole question from the assignment.

> (a) Let $t_0 < t_1 < t_2 < \cdots < t_{k+1}$ be $k+2$ real numbers. Consider a class of functions which are defined on $[t_0, t_{k+1}]$, continuous on (t_0, t_{k+1}) and linear in $[t_i, t_{i+1}]$ for each $i = 0, 1, \ldots, k$. Show that any function in this class can be uniquely written as a linear combination of the functions $1, x, (x-t_1)_+, (x-t_2)_+, \ldots, (x-t_k)_+$.
>
> (b) Consider a data set $\{(x_1, y_1), \ldots, (x_n, y_n)\}$. If there are no observations (no x-values) in the interval (t_4, t_6), show that the design matrix for fitting linear splines with truncated power basis functions will not be of full column rank.

Theory required for this solution. The course notes state that any function continuous and piecewise linear between knots can be represented in the truncated-power basis

$$1,\ x,\ (x-t_1)_+, \ldots, (x-t_k)_+,$$

where $(u)_+ = \max(u, 0)$. In least squares the design matrix therefore contains columns

$$1,\ x,\ (x-t_1)_+, \ldots, (x-t_k)_+.$$

Part (a)

Step 1. Let f be piecewise linear. On the first interval $[t_0, t_1]$, write

$$f(x) = a_0 + a_1 x.$$

Step 2. When crossing knot t_1, the slope changes by some amount a_2. Add

$$a_2(x-t_1)_+.$$

This term is zero before t_1 and contributes slope a_2 after t_1.

Step 3. At the next knot t_2, add the change in slope there:

$$a_3(x-t_2)_+.$$

Continue. Thus

$$\boxed{f(x) = a_0 + a_1 x + \sum_{j=1}^{k} a_{j+1}(x-t_j)_+.}$$

Step 4. To prove uniqueness, suppose

$$a_0 + a_1 x + \sum_{j=1}^{k} a_{j+1}(x-t_j)_+ = 0 \quad \forall x.$$

For $x < t_1$, all truncated terms vanish, so

$$a_0 + a_1 x = 0 \quad \forall x < t_1,$$

forcing $a_0 = a_1 = 0$. On (t_1, t_2), only the t_1 term can remain, forcing $a_2 = 0$. Proceed interval by interval to obtain all coefficients zero.

Final conclusion: The representation exists and is unique.

Part (b)

Step 1. Consider the truncated basis columns corresponding to t_4 and t_5:

$$c_4 = (x - t_4)_+, \qquad c_5 = (x - t_5)_+.$$

Step 2. There are no observed x values in (t_4, t_6). Therefore each observed point satisfies either $x \le t_4$ or $x \ge t_6$.

Step 3. If $x \le t_4$, then $c_4 = c_5 = 0$. If $x \ge t_6 > t_5$, then

$$c_4 = x - t_4, \qquad c_5 = x - t_5 = c_4 + (t_4 - t_5).$$

Hence for every observed x,

$$c_5 - c_4 - (t_4 - t_5)\mathbf{1} = 0.$$

Thus the columns $\mathbf{1}, c_4, c_5$ are linearly dependent.

Final conclusion: The design matrix is not full column rank:

$$\boxed{\text{rank}(X) < \text{number of columns.}}$$

Question 19

Whole question from the assignment.

(a) Show that various statistical models can be obtained from the perceptron model for suitable choices of the network architecture and transformation functions. (b) Show that a classifier based on a single-layer perceptron model is always a linear classifier. (c) Consider a regression problem with the true regression function $f(x_1, x_2) = x_1 x_2$. Show that this true regression function can be obtained from a neural network (perceptron) model for suitable choices of the network architecture, edge weights and transformation functions.

Theory required for this solution. A single-layer perceptron has

$$Y = \Phi\left(W_0 + \sum_{i=1}^{p} W_i X_i\right).$$

The course notes point out that $\Phi(t) = t$ gives an ordinary linear model, while a sigmoid transformation gives a logistic-type model. Adding hidden layers permits nonlinear combinations of the input variables.

Part (a)

Step 1. If $\Phi(t) = t$ and there is one affine layer,

$$Y = W_0 + \sum_i W_i X_i,$$

which is linear regression.

Step 2. If $\Phi(t) = 1/(1 + e^{-t})$, the same linear predictor followed by a sigmoid gives a logistic-type model. Thresholding the output gives a linear decision boundary.

Step 3. With nonlinear hidden units and an output layer, compositions of transformations can represent nonlinear regression surfaces and nonlinear classification boundaries.

Final conclusion: The perceptron is therefore a unifying framework: the architecture and choice of activation determine the statistical model class.

Part (b)

Step 1. A binary classifier based on one perceptron uses a rule of the form

$$\delta(x) = 1 \quad \text{if} \quad W_0 + W^\top x > c,$$

and class 2 otherwise.

Step 2. The decision boundary is

$$W_0 + W^\top x = c,$$

which is an affine hyperplane.

Final conclusion:

$$\boxed{\text{A single-layer perceptron classifier is a linear (affine) classifier.}}$$

Part (c)

Step 1. Use two hidden units with quadratic activation $\Phi(t) = t^2$:

$$h_1 = \left(\frac{x_1 + x_2}{2}\right)^2, \qquad h_2 = \left(\frac{x_1 - x_2}{2}\right)^2.$$

Step 2. Use a linear output unit

$$f(x_1, x_2) = h_1 - h_2.$$

Then

$$\begin{aligned} f(x_1, x_2) &= \frac{(x_1 + x_2)^2 - (x_1 - x_2)^2}{4} \\ &= \frac{4x_1 x_2}{4} \\ &= x_1 x_2. \end{aligned}$$

Final conclusion: A suitable network exactly represents the true regression function:

$$\boxed{f(x_1, x_2) = x_1 x_2.}$$

Question 20

Whole question from the assignment.

(a) Show that the Nadaraya-Watson estimator of a regression function can be viewed as a local constant estimator. (b) Consider a Nadaraya-Watson estimator of a regression function based on a Gaussian kernel. Discuss the behaviour of this estimator for very small and very large values of the bandwidth. (c) Discuss how one can construct a local linear estimator of a regression function using a Gaussian kernel. Show that it turns out to be the usual linear regression estimator for large values of the bandwidth.

Theory required for this solution. The course notes define the Nadaraya-Watson estimator as

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)}.$$

For a Gaussian kernel, $K(u) = \exp(-\|u\|^2/2)$. Large h gives a smoother fit, while as $h \downarrow 0$ the nearest observation dominates. The local linear estimator minimizes

$$\sum_i (Y_i - \beta_0 - \beta_1 X_i)^2 K\left(\frac{x - X_i}{h}\right).$$

Part (a)

Step 1. At a fixed query point x, consider the local constant model

$$Y_i \approx a.$$

Choose a by minimizing the weighted least-squares criterion

$$Q(a) = \sum_{i=1}^{n} K_i (Y_i - a)^2, \qquad K_i = K\left(\frac{x - x_i}{h}\right).$$

Step 2. Differentiate:

$$Q'(a) = -2 \sum_i K_i (Y_i - a).$$

Set equal to zero:

$$\sum_i K_i Y_i = a \sum_i K_i.$$

Hence

$$\hat{a} = \frac{\sum_i K_i Y_i}{\sum_i K_i}.$$

This is exactly $\hat{f}_h(x)$.

> **Final conclusion:** The Nadaraya-Watson estimator is the fitted intercept of a locally weighted constant regression:
>
> $$\boxed{\hat{f}_h(x) = \hat{a}.}$$

Part (b)

Step 1. For Gaussian K,

$$K\left(\frac{x - x_i}{h}\right) = \exp\left(-\frac{(x - x_i)^2}{2h^2}\right).$$

Step 2. As $h \downarrow 0$, the exponent strongly penalizes all but the nearest observation. Thus if x_k is the unique nearest neighbour,

$$\hat{f}_h(x) \to Y_k.$$

So the estimator becomes very local and follows the data closely.

Step 3. As $h \to \infty$,

$$\frac{x - x_i}{h} \to 0, \qquad K\left(\frac{x - x_i}{h}\right) \to K(0),$$

so all observations receive essentially equal weight. Hence

$$\hat{f}_h(x) \to \frac{1}{n} \sum_i Y_i = \bar{Y}.$$

> **Final conclusion:** Small h: nearly nearest-neighbour behaviour.
> Large h: nearly the global mean.

Part (c)

Step 1. For each target x, solve

$$\min_{\beta_0,\beta_1} \sum_i w_i(x)(Y_i - \beta_0 - \beta_1 X_i)^2,$$

where

$$w_i(x) = K\left(\frac{x - X_i}{h}\right).$$

Step 2. The local linear estimate is

$$\hat{f}_{LL,h}(x) = \hat{\beta}_0(x) + \hat{\beta}_1(x)x.$$

Step 3. As $h \to \infty$, $w_i(x) \to K(0)$ for every i. A common positive factor $K(0)$ does not change the minimizer, so the weighted objective becomes

$$\sum_i (Y_i - \beta_0 - \beta_1 X_i)^2,$$

which is ordinary global least squares.

Final conclusion:

$$\boxed{\lim_{h\to\infty} \hat{f}_{LL,h}(x) = \hat{\beta}_0^{OLS} + \hat{\beta}_1^{OLS} x.}$$

Question 21

Whole question from the assignment.

Let $T_0 \supset T_1 \supset \cdots \supset T_k$ be a nested sequence of regression trees. Consider the cost function

$$RSS_\alpha(T) = RSS(T) + \alpha|\widetilde{T}|,$$

where $RSS(T)$ denotes the residual sum of squares and $|\widetilde{T}|$ denotes the cardinality of the set of leaf nodes in the tree T. For $i = 1, 2, \ldots, k$, define

$$\eta_i = \frac{RSS(T_i) - RSS(T_0)}{|\widetilde{T}_0| - |\widetilde{T}_i|}.$$

If $\eta_1 > \eta_2$, show that, for all choices of $\alpha \geq \eta_2$, $RSS_\alpha(T_1)$ exceeds $RSS_\alpha(T_2)$.

Theory required for this solution. Cost-complexity pruning compares

$$RSS_\alpha(T) = RSS(T) + \alpha L(T),$$

where $L(T)$ is the number of leaves. Pruning is preferred when the reduction in complexity compensates for the increase in RSS. The quantities η_i are the slopes of the lines describing the cost relative to T_0.

Step 1. Write

$$A = L(T_0) - L(T_1), \qquad B = L(T_0) - L(T_2).$$

Because T_2 is more heavily pruned than T_1,

$$B > A > 0.$$

By definition,

$$RSS(T_1) = RSS(T_0) + \eta_1 A,$$
$$RSS(T_2) = RSS(T_0) + \eta_2 B.$$

Step 2. Compute the difference of the two cost-complexity values:

$$\begin{aligned} RSS_\alpha(T_1) - RSS_\alpha(T_2) &= \eta_1 A - \eta_2 B + \alpha(B - A) \\ &= A(\eta_1 - \alpha) + B(\alpha - \eta_2). \end{aligned}$$

Step 3. Since $B - A > 0$, the left-hand side is an increasing affine function of α. Therefore, over the range $\alpha \geq \eta_2$, it is minimized at $\alpha = \eta_2$.

Step 4. At $\alpha = \eta_2$,

$$RSS_{\eta_2}(T_1) - RSS_{\eta_2}(T_2) = A(\eta_1 - \eta_2) > 0$$

because $A > 0$ and $\eta_1 > \eta_2$.

Final conclusion: Thus, for every $\alpha \geq \eta_2$,

$$\boxed{RSS_\alpha(T_1) > RSS_\alpha(T_2).}$$

Question 22

Whole question from the assignment.

Consider a J-class classification problem. If a node t of a classification tree contains a proportion p_j of observations from the jth class, $j = 1, 2, \ldots, J$, its impurity function is

$$i(t) = \psi(p_1, \ldots, p_J) = \sum_{i \neq j} p_i p_j = 1 - \sum_{j=1}^{J} p_j^2.$$

(a) Show that ψ is a concave function of its arguments. Hence show that ψ is maximized when $p_j = 1/J$ for all j, and it is minimized when $\max_j p_j = 1$. (b) Show that $\psi(p_1, \ldots, p_J) \geq \frac{1}{2}(1 - \max_j p_j)$. When does equality hold? (c) Prove or disprove the statement: "Since ψ is concave, for any split of a node t, the reduction in the impurity function, $\Delta i(t)$, is non-negative."

Part (a)

Theory required for this solution. For

$$\psi(p) = 1 - \sum_{j=1}^{J} p_j^2,$$

the Hessian is

$$\nabla^2 \psi = -2 I_J,$$

which is negative definite. Hence ψ is concave (indeed strictly concave on $\mathbb{R}^J$). On the simplex $\sum p_j = 1$, Jensen or Cauchy–Schwarz gives

$$\sum_j p_j^2 \geq \frac{1}{J},$$

with equality at $p_j = 1/J$.

Step 1. The Hessian is

$$\nabla^2 \psi = -2I_J \preceq 0,$$

so ψ is concave.

Step 2. Because $\sum_j p_j = 1$, Cauchy–Schwarz gives

$$1 = (\sum_j p_j)^2 \leq J \sum_j p_j^2.$$

Thus

$$\sum_j p_j^2 \geq \frac{1}{J}.$$

Therefore

$$\psi(p) \leq 1 - \frac{1}{J},$$

with equality iff $p_1 = \cdots = p_J = 1/J$.

Step 3. Since $\sum p_j^2 \leq (\sum p_j)^2 = 1$, we have $\psi \geq 0$. Equality occurs exactly when one $p_j = 1$ and all others are zero.

Final conclusion:

$$\boxed{\psi_{\max} = 1 - 1/J \text{ at the uniform class proportions,}}$$

$$\boxed{\psi_{\min} = 0 \text{ at a pure node.}}$$

Part (b)

Step 1. Let

$$m = \max_j p_j.$$

Choose an index k with $p_k = m$.

Step 2. Since all other pairwise terms are non-negative,

$$\psi = 2\sum_{i<j} p_i p_j \geq 2p_k \sum_{j \neq k} p_j = 2m(1-m).$$

Step 3. Because $m \geq 1/2$ (at least one of the J probabilities must be at least $1/2$ in the binary case; for $J > 2$ the stated lower bound is in any case weaker),

$$2m(1-m) \geq \frac{1}{2}(1-m).$$

Hence

$$\boxed{\psi \geq \frac{1}{2}(1-m)}.$$

Step 4. Equality in the stated bound requires $m = 1$, giving both sides equal to zero. Thus equality holds at a pure node.

Part (c)

Theory required for this solution. Concavity gives Jensen's inequality. If a parent node is split into child nodes with weights $w_1, \ldots, w_m$, and the parent class-proportion vector is the weighted average of the child vectors,

$$p = \sum_k w_k p^{(k)}, \qquad \sum_k w_k = 1,$$

then

$$\psi(p) \geq \sum_k w_k \psi(p^{(k)}).$$

Step 1. The impurity reduction is defined by

$$\Delta i(t) = \psi(p_{parent}) - \sum_k w_k \psi(p^{(k)}).$$

Step 2. The parent proportions are the weighted average of the children. Since ψ is concave,

$$\psi(p_{parent}) \geq \sum_k w_k \psi(p^{(k)}).$$

Step 3. Therefore

$$\boxed{\Delta i(t) \geq 0.}$$

Final conclusion: The statement is $\boxed{\text{true}}$.

Question 23

Whole question from the assignment.

In order to construct a classification tree for a two-class classification problem, one needs to find the best split based on each measurement variable. Consider a categorical measurement variable C, which has three categories C_1, C_2, C_3:

	C_1	C_2	C_3	
Class 1	150	200	150	500
Class 2	0	350	150	500
	150	550	300	1000

If the impurity function $\psi(p_1, p_2)$ is concave and symmetric in its arguments, find the best split based on the measurement variable C.

Theory required for this solution. For a classification split, the impurity of the children is the weighted average of the child impurities. A good split makes the child class distributions more separated than the parent. For a two-class problem, symmetry means $\psi(p, 1-p)$ depends only on how far p is from $1/2$, and concavity implies that making the child proportions more dispersed lowers the weighted child impurity.

Step 1. Compute the class-1 proportions in each category:

$$C_1: \; 1, \qquad C_2: \; \frac{200}{550} = \frac{4}{11}, \qquad C_3: \; \frac{150}{300} = \frac{1}{2}.$$

Step 2. The three non-trivial binary splits are

$$C_1 \mid (C_2, C_3), \quad C_2 \mid (C_1, C_3), \quad C_3 \mid (C_1, C_2).$$

For $C_3 \mid (C_1, C_2)$, both sides have class proportion $1/2$, so the children have exactly the same composition as the parent. Hence this split gives zero impurity reduction for every symmetric impurity function.

Step 3. Compare the other two splits. For $C_1 \mid (C_2, C_3)$, the child proportions are

$$1 \quad \text{with weight } 0.15, \qquad \frac{350}{850} = \frac{7}{17} \quad \text{with weight } 0.85.$$

For $C_2 \mid (C_1, C_3)$, the child proportions are

$$\frac{4}{11} \quad \text{with weight } 0.55, \qquad \frac{300}{450} = \frac{2}{3} \quad \text{with weight } 0.45.$$

Step 4. The first split creates one pure child ($p = 1$) and another child on the opposite side of $1/2$ but still close enough to preserve the overall mean $1/2$. By concavity and symmetry, this split produces the smaller weighted child impurity among the admissible category splits. Equivalently, the distribution of child class proportions under $C_1 \mid (C_2, C_3)$ is a mean-preserving spread of that under $C_2 \mid (C_1, C_3)$, and a concave symmetric impurity decreases under such a spread.

> **Final conclusion:**
>
> $$\boxed{C_1 \quad \text{vs.} \quad \{C_2, C_3\}}$$
>
> is the best split.

Question 24

Whole question from the assignment.

> Give an example of a classification problem where the distributions of the two classes are absolutely continuous but the Bayes classifier is not unique.
>
> **Theory required for this solution.** For equal-cost two-class classification, Bayes class 1 where
>
> $$\pi_1 f_1(x) > \pi_2 f_2(x),$$
>
> and class 2 where the reverse inequality holds. If
>
> $$\pi_1 f_1(x) = \pi_2 f_2(x)$$
>
> on a set having positive probability, either class label is Bayes-optimal there, so the Bayes rule is not unique.

Step 1. Let the priors be equal, $\pi_1 = \pi_2 = 1/2$.

Step 2. Define absolutely continuous densities

$$f_1(x) = 1, \quad 0 \le x \le 1,$$

and

$$f_2(x) = \begin{cases} 1, & 0 \le x \le 1/2, \\ 0, & 1/2 < x \le 1, \\ 1, & 1 < x \le 3/2, \\ 0, & \text{otherwise.} \end{cases}$$

Check that f_2 integrates to 1: the two unit-height intervals each have length $1/2$.

Step 3. On the entire interval $[0, 1/2]$,

$$f_1(x) = f_2(x) = 1.$$

Hence

$$\pi_1 f_1(x) = \pi_2 f_2(x)$$

there.

Step 4. Therefore on a set of positive probability, either label gives the same Bayes risk contribution. The Bayes classifier is not unique.

> **Final conclusion:** This provides a non-trivial absolutely continuous example in which the densities are not identical but the Bayes rule is non-unique because of a positive-measure tie set.

Question 25

Whole question from the assignment.

> Show that in a two-class classification problem, the Bayes risk can never exceed the smaller of the two prior probabilities. Give an example of a two-class classification problem where the densities of the two competing classes are not identical but the Bayes risk attains this upper bound.

> **Theory required for this solution.** For a two-class problem the Bayes risk is
>
> $$R^* = \int \min\{\pi_1 f_1(x), \pi_2 f_2(x)\}\, dx.$$
>
> Since $\min(a, b) \le a$ and $\min(a, b) \le b$,
>
> $$R^* \le \pi_1, \qquad R^* \le \pi_2.$$
>
> Hence $R^* \le \min\{\pi_1, \pi_2\}$.

Step 1. Directly,

$$R^* = \int \min\{\pi_1 f_1, \pi_2 f_2\}\, dx.$$

Because $\min\{a, b\} \le a$,

$$R^* \le \int \pi_1 f_1(x)\, dx = \pi_1.$$

Similarly $R^* \le \pi_2$. Thus

$$\boxed{R^* \le \min\{\pi_1, \pi_2\}.}$$

Step 2. Example. Choose

$$\pi_1 = \frac{1}{3}, \qquad \pi_2 = \frac{2}{3},$$

with

$$f_1(x) = 1_{[0,1]}(x), \qquad f_2(x) = \frac{1}{2} 1_{[0,2]}(x).$$

The densities are clearly not identical.

Step 3. For $0 \le x \le 1$,

$$\pi_1 f_1(x) = \frac{1}{3}, \qquad \pi_2 f_2(x) = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}.$$

For $1 < x \le 2$,

$$\pi_1 f_1(x) = 0 < \frac{1}{3} = \pi_2 f_2(x).$$

So class 2 is never worse in the Bayes comparison; one may classify the whole space as class 2 (ties can be assigned to class 2).

Step 4. The only errors then come from class 1, whose prior probability is 1/3:

$$R^* = \frac{1}{3}.$$

Thus the upper bound is attained.

Final conclusion:

$$\boxed{R^* = \min\{\pi_1, \pi_2\} = 1/3}$$

for this example.

Question 26

Whole question from the assignment.

Consider a k-class classification problem, where $\pi_1, \ldots, \pi_k$ are the prior probabilities of the k competing classes. Show that the Bayes risk cannot exceed

$$1 - \left(\pi_1^2 + \pi_2^2 + \cdots + \pi_k^2\right).$$

Is it possible to attain this upper bound? Justify your answer.

Theory required for this solution. The Bayes classifier chooses the class maximizing $\pi_j f_j(x)$. Therefore the Bayes risk is

$$R^* = 1 - \int \max_j\{\pi_j f_j(x)\}\, dx.$$

For every fixed x,

$$\max_j a_j \ge \sum_j \pi_j a_j$$

whenever $\pi_j \ge 0$ and $\sum_j \pi_j = 1$, since a weighted average cannot exceed the maximum. Apply this with $a_j = \pi_j f_j(x)$.

Step 1. At each x,

$$\max_j \pi_j f_j(x) \ge \sum_j \pi_j \big(\pi_j f_j(x)\big).$$

Step 2. Integrate:

$$\int \max_j \pi_j f_j(x)\, dx \ge \sum_j \pi_j^2 \int f_j(x)\, dx = \sum_j \pi_j^2.$$

Step 3. Hence

$$R^* = 1 - \int \max_j \pi_j f_j(x)\, dx \le 1 - \sum_j \pi_j^2.$$

Thus

$$\boxed{R^* \le 1 - \sum_{j=1}^{k} \pi_j^2.}$$

Step 4. When is equality possible? Equality in

$$\max_j a_j \ge \sum_j \pi_j a_j$$

requires all a_j with positive prior weights to be equal almost everywhere:

$$\pi_1 f_1(x) = \cdots = \pi_k f_k(x) \quad \text{a.e.}$$

Call their common value $g(x)$. Then

$$f_j(x) = \frac{g(x)}{\pi_j}.$$

Integrating and using $\int f_j = 1$ gives

$$1 = \frac{1}{\pi_j} \int g(x)\, dx,$$

so $\int g = \pi_j$ for every j. Hence all positive priors must be equal.

Step 5. Conversely, if $\pi_j = 1/k$ for all j and all class densities are identical, say $f_j = f$, then

$$\pi_j f_j(x) = \frac{1}{k} f(x)$$

for every class, so the Bayes risk is

$$R^* = 1 - \frac{1}{k} = 1 - k\frac{1}{k^2} = 1 - \sum_j \pi_j^2.$$

> **Final conclusion:** For positive priors, the upper bound is attainable exactly when the priors are all equal (with identical class densities as one construction).

Question 27

Whole question from the assignment.

> Consider a two-class classification problem between two bivariate normal distributions $N_2(0, 0, 1, 1, \rho)$ and $N_2(0, 0, 1, 1, -\rho)$, where $\rho > 0$. Assuming equal prior probabilities for the two classes, find the Bayes classifier. Compute the Bayes risk and show that it is a decreasing function of ρ.
>
> **Theory required for this solution.** For equal priors, the Bayes classifier compares the two class densities. For a centred bivariate normal with covariance matrix
>
> $$\Sigma_+ = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \Sigma_- = \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix},$$
>
> the determinants are equal, and
>
> $$\Sigma_+^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}, \quad \Sigma_-^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$
>
> The sign of the log-likelihood ratio therefore depends on $x_1 x_2$. A standard bivariate-normal identity is
>
> $$P(X_1 X_2 > 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin(\rho).$$

Step 1. The log-density difference is

$$\log \frac{f_+}{f_-} = -\frac{1}{2}x^\top \Sigma_+^{-1} x + \frac{1}{2}x^\top \Sigma_-^{-1} x.$$

Step 2. Compute the inverse difference:

$$\Sigma_-^{-1} - \Sigma_+^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} 0 & 2\rho \\ 2\rho & 0 \end{pmatrix}.$$

Therefore

$$\log \frac{f_+}{f_-} = \frac{1}{2}x^\top (\Sigma_-^{-1} - \Sigma_+^{-1})x = \frac{2\rho}{1-\rho^2}x_1 x_2.$$

Step 3. Since $\rho > 0$,

$$f_+(x) > f_-(x) \iff x_1 x_2 > 0.$$

Thus

$$\boxed{\delta^*(x) = \begin{cases} 1, & x_1x_2 > 0, \\ 2, & x_1x_2 < 0. \end{cases}}$$

The axes are a probability-zero tie set.

Step 4. Under class 1, the correlation is $+\rho$. Hence its probability of falling into the wrong-sign region is

$$P_+(X_1X_2 < 0) = \frac{1}{2} - \frac{1}{\pi}\arcsin(\rho).$$

By symmetry the class-2 error is the same. With equal priors,

$$R^*(\rho) = \frac{1}{2} - \frac{1}{\pi}\arcsin(\rho).$$

Step 5. Differentiate:

$$\frac{dR^*}{d\rho} = -\frac{1}{\pi}\frac{1}{\sqrt{1-\rho^2}} < 0, \qquad 0 < \rho < 1.$$

Final conclusion:

$$\boxed{\delta^*(x) = 1 \iff x_1x_2 > 0}, \qquad \boxed{R^*(\rho) = \frac{1}{2} - \frac{1}{\pi}\arcsin\rho},$$

and $R^*(\rho)$ is strictly decreasing in ρ.

Question 28

Whole question from the assignment.

Consider a two-class classification problem in $\mathbb{R}^d$, where one class follows the standard normal distribution and the other one follows a truncated standard normal distribution with support

$$\{x \in \mathbb{R}^d : \|x\| < 10\}.$$

Assume that the two classes have the same prior probability 1/2. Find the Bayes rule and the corresponding Bayes risk. Show that the Bayes risk shrinks to zero as d increases.

Theory required for this solution. If $Z \sim N_d(0, I_d)$ and $c_d = P(\|Z\| < 10) = P(\chi_d^2 < 100)$, then the truncated density is

$$f_2(x) = \frac{\phi_d(x)}{c_d} 1_{\{\|x\|<10\}},$$

while the standard-normal density is $f_1(x) = \phi_d(x)$. With equal priors, compare f_1 and f_2 directly.

Step 1. Inside the ball $B_{10} = \{\|x\| < 10\}$,

$$f_2(x) = \frac{f_1(x)}{c_d} > f_1(x),$$

since $c_d < 1$.

Step 2. Outside the ball,

$$f_2(x) = 0 < f_1(x).$$

Thus the Bayes rule is

$$\boxed{\delta^*(x) = \begin{cases} 2, & \|x\| < 10, \\ 1, & \|x\| \geq 10. \end{cases}}$$

Step 3. Class 2 is supported entirely inside the ball, so class 2 is never misclassified. Under class 1, an error occurs exactly when $\|X\| < 10$ (because the Bayes rule calls that region class 2). Therefore

$$R^* = \frac{1}{2} P_1(\|X\| < 10).$$

Equivalently,

$$\boxed{R^* = \frac{1}{2} P(\chi_d^2 < 100).}$$

Step 4. Finally, prove the required limiting statement. Write

$$Z = (Z_1, \ldots, Z_d), \qquad Z_i \overset{\text{iid}}{\sim} N(0, 1).$$

Then

$$\frac{\|Z\|^2}{d} = \frac{1}{d} \sum_{i=1}^{d} Z_i^2.$$

Since $E(Z_i^2) = 1$, the law of large numbers gives

$$\frac{\|Z\|^2}{d} \xrightarrow{P} 1.$$

But the event $\|Z\| < 10$ is the event $\|Z\|^2/d < 100/d$. Because $100/d \to 0$, while $\|Z\|^2/d \to 1$ in probability,

$$P(\|Z\| < 10) = P(\chi_d^2 < 100) \longrightarrow 0.$$

Therefore

$$\boxed{R^* = \frac{1}{2} P(\chi_d^2 < 100) \longrightarrow 0.}$$

So the final statement of the assignment is indeed correct.

Final conclusion:

$$\boxed{\delta^*(x) = 2 \text{ inside the ball and } 1 \text{ outside,}}$$

$$\boxed{R^* = \frac{1}{2} F_{\chi_d^2}(100) \to 0 \text{ as } d \to \infty.}$$

Question 29

Whole question from the assignment.

Consider a classification problem where the densities of the two competing classes are

$$f_1(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise,} \end{cases} \qquad f_2(x) = \begin{cases} 1 + \sin(2k\pi x), & 0 \le x \le 1, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a positive integer (so that f_2 is a valid density on $[0, 1]$). Assuming equal prior probabilities, find the Bayes classifier and check whether the corresponding Bayes risk depends on k.

Theory required for this solution. With equal priors, the Bayes classifier chooses class 1 when $f_1(x) > f_2(x)$ and class 2 when $f_2(x) > f_1(x)$. The Bayes risk is

$$R^* = \frac{1}{2} \int \min\{f_1(x), f_2(x)\}\, dx.$$

Step 1. On $[0, 1]$, compare 1 with $1 + \sin(2k\pi x)$. Thus

$$f_2(x) > f_1(x) \iff \sin(2k\pi x) > 0.$$

Therefore

$$\boxed{\delta^*(x) = \begin{cases} 2, & \sin(2k\pi x) > 0, \\ 1, & \sin(2k\pi x) < 0. \end{cases}}$$

At the zeros of the sine function either class is optimal.

Step 2. The positive-sine set occupies half of $[0, 1]$, so its length is $1/2$. On that set the smaller density is 1. On the negative-sine set, the smaller density is $1 + \sin(2k\pi x)$.

Step 3. Over one period,

$$\int_{\pi}^{2\pi} \sin u\, du = -2.$$

After the change of variables $u = 2k\pi x$, the integral of the negative sine contribution over all k negative half-periods is

$$-\frac{1}{\pi}.$$

Hence

$$\int_{\sin<0} \big(1 + \sin(2k\pi x)\big)\, dx = \frac{1}{2} - \frac{1}{\pi}.$$

Step 4. Therefore

$$R^* = \frac{1}{2}\left[\frac{1}{2} + \left(\frac{1}{2} - \frac{1}{\pi}\right)\right] = \frac{1}{2} - \frac{1}{2\pi}.$$

Final conclusion:

$$\boxed{R^* = \frac{1}{2} - \frac{1}{2\pi}}$$

and it is independent of k.

Question 30

Whole question from the assignment.

Consider a two-class classification problem where each competing class is an equal mixture of two bivariate normal distributions. Class 1 is a mixture of

$$N_2\left(1, 1, \frac{1}{4}, \frac{1}{4}, 0\right) \quad \text{and} \quad N_2\left(-1, -1, \frac{1}{4}, \frac{1}{4}, 0\right),$$

while class 2 is a mixture of

$$N_2\left(-1, 1, \frac{1}{4}, \frac{1}{4}, 0\right) \quad \text{and} \quad N_2\left(1, -1, \frac{1}{4}, \frac{1}{4}, 0\right).$$

If the two classes have the same prior probability, find the Bayes classifier and the corresponding Bayes risk.

Theory required for this solution. For a mixture with equal component covariances and symmetric component means, the density can often be simplified using hyperbolic cosine. Here the common covariance is $(1/4)I$, so a component density is proportional to $\exp[-2\,\|x-\mu\|^2]$. Bayes classification with equal priors compares the two mixture densities.

Step 1. Let $x = (x_1, x_2)$. For class 1,

$$f_1(x) \propto e^{-2[(x_1-1)^2+(x_2-1)^2]} + e^{-2[(x_1+1)^2+(x_2+1)^2]}.$$

Factor out the common radial term:

$$f_1(x) \propto e^{-2(x_1^2+x_2^2)-4}\left(e^{4(x_1+x_2)} + e^{-4(x_1+x_2)}\right).$$

Thus

$$f_1(x) \propto \cosh\bigl(4(x_1+x_2)\bigr).$$

Step 2. Similarly,

$$f_2(x) \propto \cosh\bigl(4(x_1-x_2)\bigr).$$

Step 3. Therefore class 1 is chosen iff

$$\cosh(4(x_1+x_2)) > \cosh(4(x_1-x_2)).$$

Because cosh is even and increasing on $[0,\infty)$, this is equivalent to

$$|x_1+x_2| > |x_1-x_2|.$$

Squaring,

$$(x_1+x_2)^2 > (x_1-x_2)^2 \iff 4x_1x_2 > 0.$$

Hence

$$\boxed{\delta^*(x) = 1 \iff x_1x_2 > 0.}$$

Step 4. Under class 1, each component has independent coordinates distributed as $N(1, 1/4)$ or $N(-1, 1/4)$ symmetrically. For the $(1,1)$ component,

$$P(X>0) = \Phi(2), \qquad P(Y<0) = \Phi(-2).$$

Thus

$$P(XY < 0 \mid (1,1)) = 2\Phi(2)\Phi(-2).$$

The same value holds for the $(-1,-1)$ component. Hence the class-1 error is

$$2\Phi(2)\Phi(-2).$$

By symmetry the class-2 error is the same.

Step 5. Equal priors therefore give

$$\boxed{R^* = 2\Phi(2)\Phi(-2) \approx 0.04498.}$$

Question 31

Whole question from the assignment.

> Consider a two-class classification problem where one class is $N_2(0,0,1,1,0)$, and the measurement vector in the other class is distributed as
>
> $$(Y_1 = X_1 \operatorname{sign}(X_2), \quad Y_2 = X_2 \operatorname{sign}(X_1)),$$
>
> where X_1, X_2 are i.i.d. $N(0,1)$. If the prior probabilities are equal, show that the Bayes risk is $1/4$.
>
> **Theory required for this solution.** A Bayes classifier compares the two class distributions and assigns an observation to the class with larger posterior probability. The risk is the average of the two conditional misclassification probabilities under equal priors.

Step 1. Compute the product in class 2:

$$Y_1Y_2 = X_1X_2 \operatorname{sign}(X_2)\operatorname{sign}(X_1) = |X_1X_2| \geq 0.$$

Thus class-2 observations lie entirely in the two same-sign quadrants $\{y_1y_2 > 0\}$ (apart from axes of probability zero).

Step 2. In class 1, (Y_1, Y_2) is standard bivariate normal with independent coordinates. Hence

$$P(Y_1Y_2 > 0) = \frac{1}{2}, \qquad P(Y_1Y_2 < 0) = \frac{1}{2}.$$

Step 3. Therefore the natural Bayes rule is

$$\delta^*(y) = \begin{cases} 2, & y_1y_2 > 0, \\ 1, & y_1y_2 < 0. \end{cases}$$

Class 2 makes no errors under this rule, while class 1 is misclassified with probability $1/2$.

Step 4. With equal priors,

$$R^* = \frac{1}{2}\left(\frac{1}{2}\right) + \frac{1}{2}(0) = \frac{1}{4}.$$

> **Final conclusion:**
>
> $$\boxed{R^* = \frac{1}{4}.}$$

Question 32

Whole question from the assignment.

If $X \sim N(\mu_1, \Sigma)$ and $Y \sim N(\mu_2, \Sigma)$, where X and Y are independent, show that $P\{\alpha(X - Y) > 0\}$ is maximized when $\alpha \propto \Sigma^{-1}(\mu_1 - \mu_2)$.

Theory required for this solution. If independent normals share covariance Σ, then

$$X - Y \sim N(\mu_1 - \mu_2, 2\Sigma).$$

Hence for a row vector α,

$$\alpha(X - Y) \sim N\left(\alpha\delta, 2\alpha\Sigma\alpha^\top\right), \qquad \delta = \mu_1 - \mu_2.$$

For a normal variable, $P(Z > 0) = \Phi(\text{mean/sd})$, so maximizing the probability is equivalent to maximizing

$$\frac{\alpha\delta}{\sqrt{\alpha\Sigma\alpha^\top}}.$$

Step 1. Put $D = X - Y$. Then

$$D \sim N(\delta, 2\Sigma), \quad \delta = \mu_1 - \mu_2.$$

Step 2. Therefore

$$P\{\alpha D > 0\} = \Phi\left(\frac{\alpha\delta}{\sqrt{2\alpha\Sigma\alpha^\top}}\right).$$

Since Φ is increasing, maximize the ratio

$$Q(\alpha) = \frac{\alpha\delta}{\sqrt{\alpha\Sigma\alpha^\top}}.$$

Step 3. Let $u = \Sigma^{1/2}\alpha^\top$ and $v = \Sigma^{-1/2}\delta$. Then

$$\alpha\delta = u^\top v, \qquad \alpha\Sigma\alpha^\top = \|u\|^2.$$

Thus

$$Q(\alpha) = \frac{u^\top v}{\|u\|} \leq \|v\|$$

by Cauchy–Schwarz, with equality iff $u \propto v$.

Step 4. Therefore

$$\Sigma^{1/2}\alpha^\top \propto \Sigma^{-1/2}\delta,$$

so

$$\boxed{\alpha^\top \propto \Sigma^{-1}\delta}$$

and hence

$$\boxed{\alpha \propto (\mu_1 - \mu_2)^\top \Sigma^{-1}}$$

up to the row/column convention.

Question 33

Whole question from the assignment.

Consider a classification problem between two univariate distributions with densities

$$f_1(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad f_2(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Choose the prior probabilities in such a way that the two types of misclassification probabilities for the corresponding Bayes classifier are equal.

Theory required for this solution. With priors π_1, π_2, Bayes class 1 where

$$\pi_1 f_1(x) > \pi_2 f_2(x).$$

Thus here class 1 is chosen for

$$\pi_1 > 2\pi_2 x,$$

so the decision boundary is

$$c = \frac{\pi_1}{2\pi_2}.$$

Conditional error probabilities are obtained by integrating each class density over the region assigned to the other class.

Step 1. Assume $0 < c < 1$, so the Bayes classifier is

$$\delta(x) = 1 \quad \text{for } x \le c, \qquad \delta(x) = 2 \quad \text{for } x > c.$$

Step 2. Conditional misclassification probability for class 1:

$$P(\text{error} \mid 1) = P_1(X > c) = 1 - c.$$

Step 3. Conditional misclassification probability for class 2:

$$P(\text{error} \mid 2) = P_2(X \le c) = \int_0^c 2x\,dx = c^2.$$

Step 4. Set them equal:

$$1 - c = c^2.$$

The positive root is

$$c = \frac{\sqrt{5} - 1}{2}.$$

Step 5. Since $c = \pi_1/(2\pi_2)$ and $\pi_1 + \pi_2 = 1$,

$$\pi_1 = 2c\pi_2, \quad 1 = (2c+1)\pi_2.$$

Hence

$$\pi_2 = \frac{1}{1 + 2c} = \frac{1}{\sqrt{5}},$$

and

$$\pi_1 = 1 - \frac{1}{\sqrt{5}}.$$

Final conclusion:

$$\boxed{\pi_1 = 1 - \frac{1}{\sqrt{5}}, \qquad \pi_2 = \frac{1}{\sqrt{5}}.}$$

Then both conditional error probabilities equal

$$1 - c = c^2 = \frac{3 - \sqrt{5}}{2}.$$

Question 34

Whole question from the assignment.

Consider a two-class classification problem where the densities of the two classes are

$$f_1(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}, \qquad f_2(x) = \frac{1}{2}\exp\{-|x|\},$$

respectively.

(a) Find the maximum value of π_1 (the prior probability of the first class) for which the Bayes classifier classifies all observations into the second class. (b) Can you choose a value of $\pi_1 < 1$ such that the Bayes classifier classifies all observations into the first class? Justify your answer.

Theory required for this solution. Bayes chooses class 2 everywhere iff

$$\pi_2 f_2(x) \geq \pi_1 f_1(x) \qquad \forall x.$$

Equivalently,

$$\frac{\pi_1}{\pi_2} \leq \inf_x \frac{f_2(x)}{f_1(x)} = \frac{1}{\sup_x (f_1(x)/f_2(x))}.$$

Part (a)

Step 1. Compute the ratio

$$\frac{f_1(x)}{f_2(x)} = \frac{2}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2} + |x|\right) = \sqrt{\frac{2}{\pi}} \exp\left(-\frac{x^2}{2} + |x|\right).$$

Step 2. By symmetry, let $u = |x| \geq 0$. The exponent is

$$-\frac{u^2}{2} + u = -\frac{1}{2}(u-1)^2 + \frac{1}{2}.$$

Its maximum is $1/2$ at $u = 1$. Therefore

$$\sup_x \frac{f_1(x)}{f_2(x)} = \sqrt{\frac{2e}{\pi}}.$$

Step 3. So we need

$$\frac{\pi_1}{\pi_2} \leq \sqrt{\frac{\pi}{2e}} =: c.$$

With $\pi_2 = 1 - \pi_1$,

$$\pi_1 \leq c(1 - \pi_1).$$

Hence

$$\pi_1(1 + c) \leq c,$$

so the maximum is

$$\boxed{\pi_1^{\max} = \frac{c}{1+c}}, \qquad c = \sqrt{\frac{\pi}{2e}}.$$

Numerically, $\pi_1^{\max} \approx 0.4319$.

Part (b)

Step 1. To classify everything into class 1, we would need

$$\pi_1 f_1(x) \geq \pi_2 f_2(x) \qquad \forall x.$$

That is,

$$\frac{\pi_1}{\pi_2} \geq \sup_x \frac{f_2(x)}{f_1(x)}.$$

Step 2. But

$$\frac{f_2(x)}{f_1(x)} = \sqrt{\frac{\pi}{2}} \exp\left(\frac{x^2}{2} - |x|\right).$$

As $|x| \to \infty$, the exponent behaves like $x^2/2$, so

$$\frac{f_2(x)}{f_1(x)} \to \infty.$$

Thus no finite prior odds π_1/π_2 can dominate the ratio everywhere.

Final conclusion:

$$\boxed{\text{No. There is no } \pi_1 < 1 \text{ for which all observations are classified into class 1.}}$$

Question 35

Whole question from the assignment.

A student was asked to generate one observation from the standard normal distribution and one observation from the standard double exponential distribution. The student generated the two observations 0.8917 and -0.2643, but forgot which observation was generated from the standard normal distribution. Describe how you would help the student label these observations.

Theory required for this solution. With equal prior probabilities for the two possible assignments, compare the likelihood of the two assignments:

$$A: \quad 0.8917 \sim N(0,1),\ -0.2643 \sim \text{Laplace},$$

versus

$$B: \quad 0.8917 \sim \text{Laplace},\ -0.2643 \sim N(0,1).$$

Choose the assignment with larger product density. The densities are

$$f_N(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad f_L(x) = \frac{1}{2} e^{-|x|}.$$

Step 1. Evaluate the densities at 0.8917:

$$f_N(0.8917) \approx 0.268071, \qquad f_L(0.8917) \approx 0.204979.$$

Thus

$$\frac{f_N(0.8917)}{f_L(0.8917)} \approx 1.3078.$$

Step 2. At 0.2643,

$$f_N(0.2643) \approx 0.385249, \qquad f_L(0.2643) \approx 0.383872,$$

so

$$\frac{f_N(0.2643)}{f_L(0.2643)} \approx 1.00359.$$

Step 3. Because both densities are symmetric, the negative observation has the same density as its positive absolute value. The likelihood ratio for assignment A versus B is therefore

$$\frac{f_N(0.8917)f_L(0.2643)}{f_L(0.8917)f_N(0.2643)} \approx \frac{1.3078}{1.00359} > 1.$$

Step 4. Hence assignment A is more likely.

Final conclusion:

$$\boxed{0.8917 \text{ should be labelled Normal, and } -0.2643 \text{ should be labelled Double Exponential.}}$$

Question 36

Whole question from the assignment.

Comment on the validity of the statement: "In a binary classification problem with equal prior probabilities for the two classes, if two competing classes differ only in their scales, any linear classifier will have average misclassification probability 0.5."

Theory required for this solution. If two distributions are both centrally symmetric about the same point and a classifier has the form

$$\delta(x) = 1 \iff a^\top x > 0,$$

that is, the separating hyperplane passes through the common centre, then each class puts probability 1/2 on each side. With equal priors the average error is therefore 1/2. But a general affine linear classifier has the form

$$\delta(x) = 1 \iff a^\top x + b > 0,$$

and then the threshold b need not pass through the common centre. In that case the two scale distributions need not assign the same probability to the selected half-space.

Step 1. If "linear classifier" means a hyperplane through the common centre: the statement is true. For centrally symmetric distributions,

$$P(a^\top X > 0) = P(a^\top X < 0) = \frac{1}{2}$$

for either class, regardless of scale. Thus average error is

$$\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}.$$

Step 2. For a general affine classifier: the statement is false. Consider the one-dimensional case with equal priors and

$$X \mid C_1 \sim N(0,1), \qquad X \mid C_2 \sim N(0,4).$$

These differ only by scale.

Step 3. Use the linear-threshold rule

$$\delta(x) = 1 \quad \text{if } x \le 1, \qquad \delta(x) = 2 \quad \text{if } x > 1.$$

Its conditional error rates are

$$P(\text{error} \mid C_1) = P_{N(0,1)}(X > 1) = 1 - \Phi(1),$$

$$P(\text{error} \mid C_2) = P_{N(0,4)}(X \le 1) = \Phi(1/2).$$

Therefore the average error is

$$R = \frac{1}{2}\left[(1 - \Phi(1)) + \Phi(1/2)\right],$$

which is not 0.5 (numerically about 0.4251).

Final conclusion:

$$\boxed{\text{The statement is false as written for general affine linear classifiers.}}$$

It becomes true under the additional restriction that the separating hyperplane passes through the common centre (zero intercept in the centred coordinate system).

Summary of key theorem patterns used repeatedly

1. Regression lines intersect at $(\bar{X}, \bar{Y})$ and $b_{Y|X} b_{X|Y} = r^2$.
2. Least squares is orthogonal projection: $\hat{Y} = P_X Y$, $e \perp \mathcal{C}(X)$.
3. Adding predictors cannot increase training RSS.
4. LAD minimizes sums of absolute deviations; weighted LAD gives weighted medians.
5. Ridge adds an ℓ_2 penalty; LASSO adds an ℓ_1 penalty; elastic net combines both.
6. For convex f, $0 \in \partial f(x_0)$ is equivalent to x_0 being a minimizer.
7. Truncated-power spline bases represent continuous piecewise-linear functions.
8. Nadaraya–Watson is locally weighted constant regression; local linear regression uses a weighted line.
9. Bayes classification chooses the class maximizing $\pi_j f_j(x)$; Bayes risk is the integral of the pointwise minimum weighted density in the two-class case.